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PURE MATHEMATICS WITH MECHANICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH MECHANICS

CHAPTER 18

MOMENTUM AS A VECTOR

SUGGESTED TIME: 12 PERIODS

IN THIS CHAPTER

This chapter introduces linear momentum as a vector quantity and uses it to give a deeper meaning to Newton's laws of motion. Through the principle of conservation of momentum, learners analyse collisions, explosions, impulse, rocket motion and the motion of the centre of mass in one and two dimensions, with direct applications to road safety and everyday life in Cameroon.

- By the end of this chapter I can define linear momentum, state its vector nature and its SI unit, and compute the momentum of one or several bodies.
- By the end of this chapter I can relate force to the rate of change of momentum and use this as the general form of Newton's second law.
- By the end of this chapter I can state and apply the principle of conservation of linear momentum to isolated systems of interacting bodies.
- By the end of this chapter I can distinguish elastic from inelastic collisions and solve one-dimensional collision problems using conservation of momentum and, where appropriate, conservation of kinetic energy.
- By the end of this chapter I can analyse two-dimensional collisions by resolving momentum into perpendicular components.
- By the end of this chapter I can define impulse, use the impulse-momentum theorem, and interpret force-time graphs, including applications to safety devices.

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COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

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1. Momentum, Force and Impulse
2. Conservation of Linear Momentum
3. Collisions in One Dimension: Elastic and Inelastic
4. Collisions and Momentum in Two Dimensions
5. Explosions, Rockets and Centre of Mass
6. Real-Life Applications and Synthesis of Momentum as a Vector
7. Integration activity
8. Methods and worked examples
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10. Solutions to the exercises
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CHAPTER OBJECTIVES

- By the end of this chapter I can define linear momentum, state its vector nature and its SI unit, and compute the momentum of one or several bodies.
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- By the end of this chapter I can analyse two-dimensional collisions by resolving momentum into perpendicular components.
- By the end of this chapter I can define impulse, use the impulse-momentum theorem, and interpret force-time graphs, including applications to safety devices.
- By the end of this chapter I can explain rocket propulsion and explosions using conservation of momentum.
- By the end of this chapter I can locate the centre of mass of a system of particles and relate its motion to the total momentum of the system.
- By the end of this chapter I can model real-life situations (vehicle crashes, recoil of a gun, sports impacts) quantitatively using momentum as a vector.

PREREQUISITES

- Newton's three laws of motion and their use with F = ma (Lower Sixth Mechanics)
- Kinematics of motion in a straight line: velocity, acceleration, suvat equations
- Kinetic energy and the work–energy principle
- Vector algebra: components, resolution, addition of vectors, use of unit vectors i and j
- Basic trigonometry and solving simultaneous linear equations

1 MOMENTUM, FORCE AND IMPULSE

During a busy morning at the Mile 4 motor park in Limbe, a fully loaded 70-seater bus and an empty taxi collide at low speed: the taxi is thrown violently backwards while the bus barely slows down. Yet a few kilometres away, a goalkeeper at the Limbe Omnisport stadium stops a fast football with his bare hands and feels only a sting, whereas the same ball hitting a concrete wall rebounds sharply. Why does the "quantity of motion" of a body depend on both its mass and its velocity? Why do some collisions bounce and others stick? And how do engineers use these ideas to design safer vehicles and launch rockets? This chapter answers these questions by treating **momentum** as a vector quantity. We begin with the fundamental definitions: momentum, force and impulse.

1.1 Linear Momentum: Definition and Vector Nature

An observation to build intuition.

Imagine two objects moving towards you at 10 m s^{-1}: a football and a lorry. You could stop the football with your hands; the lorry would be impossible to stop. Now imagine the same lorry moving at 0.5 m s^{-1}: a few strong people could stop it. So the "quantity of motion" of a body grows with both its mass and its velocity. Moreover, a lorry moving towards you is very different from one moving away from you: direction matters. This leads us to define momentum as a vector.

DEFINITION **Linear momentum**

The **linear momentum** $\vec{p}$ of a body of mass m moving with velocity $\vec{v}$ is

$$\vec{p} = m\vec{v}.$$

Momentum is a vector quantity: it has the same direction as the velocity of the body. Its SI unit is the kilogram-metre per second, kg m s^{-1}, which is equivalent to the newton-second (N s), since 1 N = 1 kg m s^{-2}.

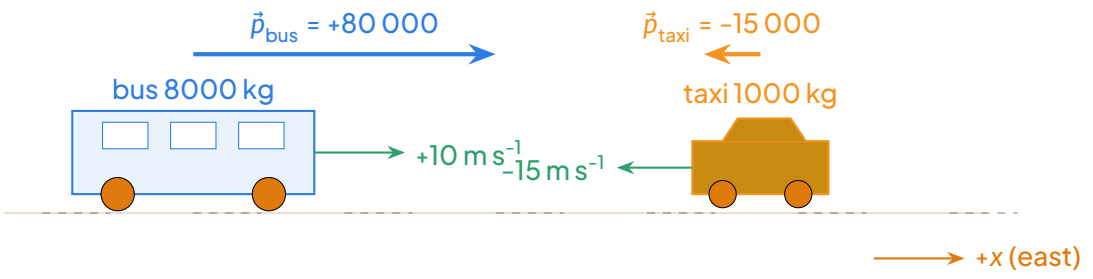
In one-dimensional problems we work with the component of momentum along a chosen positive direction: $p = mv$, where v carries a sign (+ or −). This sign is the mathematical translation of the direction of motion.

EXEMPLE **Momentum of the bus and the taxi**

A loaded bus of mass 8000 kg moves east at 10 m s^{-1}; a taxi of mass 1000 kg moves west at 15 m s^{-1}. Taking east as positive:

$$p_{\text{bus}} = 8000 \times 10 = 80\,000 \text{ kg m s}^{-1}, \qquad p_{\text{taxi}} = 1000 \times (-15) = -15\,000 \text{ kg m s}^{-1}.$$

The bus has more than five times the magnitude of momentum of the taxi, in the opposite direction. This is why, in a collision, the bus barely notices the impact while the taxi is thrown backwards.



Momentum vectors of a bus and a taxi moving in opposite directions. Arrow lengths are to scale: the bus carries far more momentum than the taxi, in the opposite direction.

1.2 Momentum of a System of Particles

DEFINITION **Total momentum of a system**

The total momentum of a system of particles is the vector sum of the individual momenta:

$$\vec{P} = \vec{p}_1 + \vec{p}_2 + \cdots + \vec{p}_n = \sum_{i=1}^{n} m_i \vec{v}_i.$$

In one dimension this vector sum becomes an algebraic sum once a positive direction is chosen.

WORKED EXAMPLE 1 **Total momentum of two vehicles**

A lorry of mass 3000 kg travels east at 12 m s^{-1}. A car of mass 900 kg travels west at 20 m s^{-1}. Find the total momentum of the system.

SOLUTION

Solution. Take east as positive.

$$P = m_1 v_1 + m_2 v_2 = 3000(12) + 900(-20)$$
$$= 36\,000 - 18\,000 = 18\,000 \text{ kg m s}^{-1}.$$

The total momentum is 18 000 kg m s^{-1} directed east (the positive sign tells us the direction).

WARNING **Momentum is a vector – mind the signs!**

The single most common error in GCE momentum questions is forgetting that a body moving in the negative direction has negative velocity and therefore negative momentum. When a ball rebounds, its velocity changes sign: if $u = +5\ \text{m s}^{-1}$ and it rebounds at $3\ \text{m s}^{-1}$, then $v = -3\ \text{m s}^{-1}$, not $+3$. Always state your positive direction before substituting numbers.

1.3 Newton's Second Law in Momentum Form

Newton originally stated his second law not as $F = ma$ but in terms of the change of "motion" (momentum). This more general form is the one required at Advanced Level.

PROPERTY **Newton's second law – general form (proof examinable)**

The resultant force acting on a body equals the rate of change of its momentum:

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(m\vec{v}).$$

Derivation of $\vec{F} = m\vec{a}$ for constant mass. If the mass m is constant, it can be taken out of the derivative:

$$\vec{F} = \frac{d}{dt}(m\vec{v}) = m\frac{d\vec{v}}{dt} = m\vec{a}.$$

Thus $F = ma$ is a special case of $F = \dfrac{dp}{dt}$, valid only when the mass does not change.

When is $F = \dfrac{dp}{dt}$ more general than $F = ma$?

If the mass varies with time, the product rule gives

$$\vec{F} = \frac{d}{dt}(m\vec{v}) = m\frac{d\vec{v}}{dt} + \vec{v}\frac{dm}{dt},$$

which is not simply $m\vec{a}$. Variable-mass systems include:

- a **rocket** burning fuel and ejecting exhaust gases (its mass decreases);
- a conveyor belt onto which sand falls at a steady rate (mass increases);
- a raindrop growing as it falls through a cloud.

For such systems, only the momentum form of Newton's second law is correct. We shall exploit this in the study of rocket motion later in the chapter.

1.4 Impulse and the Impulse–Momentum Theorem

Intuition.

To change the momentum of a body you must apply a force for some time. A small force applied for a long time can produce the same change of momentum as a large force applied briefly. The quantity that captures "force × duration" is called impulse.

DEFINITION **Impulse**

The **impulse** $\vec{J}$ of a constant force $\vec{F}$ acting for a time interval Δt is

$$\vec{J} = \vec{F}\,\Delta t.$$

For a variable force, the impulse over $[t_1, t_2]$ is

$$\vec{J} = \int_{t_1}^{t_2} \vec{F}\,dt.$$

Impulse is a vector in the direction of the force; its SI unit is the newton-second (N s), equivalent to kg m s^{-1}.

PROPERTY **Impulse–momentum theorem (proof examinable)**

The impulse of the resultant force on a body equals the change of momentum of the body:

$$\vec{J} = \Delta\vec{p} = m\vec{v} - m\vec{u},$$

where $\vec{u}$ and $\vec{v}$ are the velocities before and after the force acts. Proof. From Newton's second law, $\vec{F} = \dfrac{d\vec{p}}{dt}$. Integrating between t_1 and t_2:

$$\vec{J} = \int_{t_1}^{t_2} \vec{F}\,dt = \int_{t_1}^{t_2} \frac{d\vec{p}}{dt}\,dt = \vec{p}(t_2) - \vec{p}(t_1) = m\vec{v} - m\vec{u}. \qquad \blacksquare$$

METHOD **Solving impulse problems**

1. Choose a positive direction and state it clearly (e.g. "take the initial direction of motion as positive").
2. Write all velocities with their signs: u before, v after. A reversed velocity is negative.
3. Compute the change of momentum: $\Delta p = m(v - u)$.
4. Apply the theorem: $J = \Delta p$, and if the contact time Δt is known, the average force is
$$F_{\text{avg}} = \frac{\Delta p}{\Delta t} = \frac{m(v - u)}{\Delta t}.$$
5. Interpret the sign of J or F: it gives the direction of the impulse/force relative to your

chosen positive direction.

WORKED EXAMPLE 2 **Ball bouncing off a wall**

A ball of mass 0.5 kg moving horizontally at 8 m s^{-1} strikes a wall perpendicularly and rebounds at 6 m s^{-1}. The contact with the wall lasts 0.05 s. Find (a) the impulse on the ball; (b) the average force exerted by the wall.

SOLUTION

Solution. Take the initial direction of motion (towards the wall) as positive. Then $u = +8$ m s^{-1} and, since the ball rebounds, $v = -6$ m s^{-1}.
(a) Impulse:

$$J = m(v - u) = 0.5(-6 - 8) = 0.5 \times (-14) = -7 \text{ N s}.$$

The impulse has magnitude 7 N s, directed away from the wall.
(b) Average force:

$$F_{\text{avg}} = \frac{J}{\Delta t} = \frac{-7}{0.05} = -140 \text{ N}.$$

The wall pushes the ball with an average force of 140 N directed away from the wall. Note that had we written $v = +6$, we would have obtained the absurd answer $J = -1$ N s: the sign convention is essential.

1.5 Force–Time Graphs

In real impacts (a ball on a bat, a car in a crash test) the force is not constant: it rises from zero, peaks, and falls back to zero. Since $J = \int F\,dt$, the impulse has a beautiful graphical interpretation.

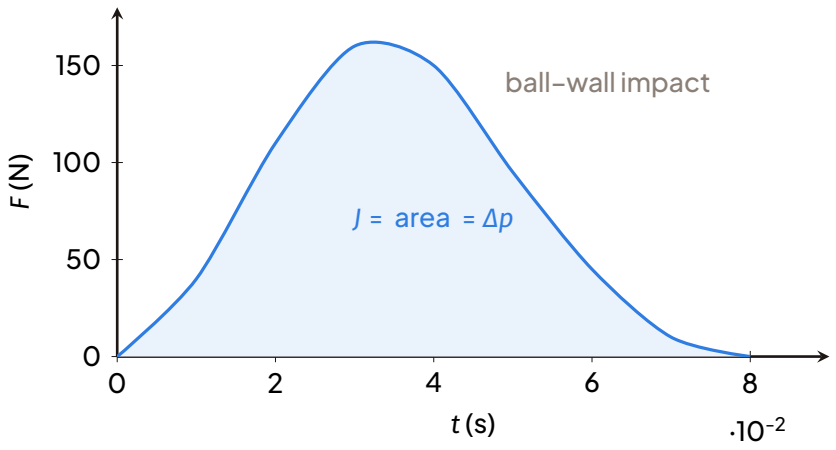
PROPERTY **Impulse as an area**

The impulse of a force over a time interval equals the area under the force–time graph between the curve and the time axis:

$$J = \int_{t_1}^{t_2} F\,dt = \text{area under the } F\text{–}t \text{ curve}.$$

The average force is then the constant force that would give the same area over the same duration:

$$F_{\text{avg}} = \frac{J}{\Delta t} = \frac{\text{area under the curve}}{\text{duration}}.$$



Force–time graph for a ball bouncing off a wall. The shaded area under the curve is the impulse, equal to the change of momentum of the ball.

EXEMPLE **Average force from a graph**

Suppose the area under the curve above is estimated (by counting squares or by trapezia) as 7.0 N s over a contact time of 0.08 s. Then

$$F_{\text{avg}} = \frac{7.0}{0.08} = 87.5\ \text{N},$$

even though the peak force reaches about 160 N. The average force is what the impulse–momentum theorem delivers directly.

DID YOU KNOW? **A frequent GCE question**

"Show that the area under a force–time graph represents the impulse" and "estimate the impulse from the given graph by counting squares" appear regularly in GCE Advanced Level papers. Practise both: the theoretical link $J = \int F\,dt = \Delta p$, and the numerical skill of estimating an irregular area (count full squares, group partial squares, or use the trapezium rule).

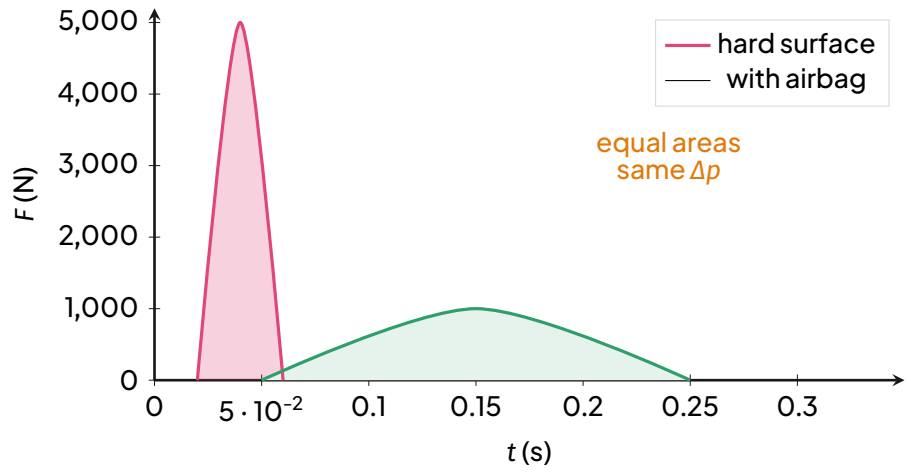
1.6 Application to Safety: Increasing Impact Time

Rewrite the impulse–momentum theorem as

$$F_{\text{avg}} = \frac{\Delta p}{\Delta t}.$$

In any given impact, the change of momentum Δp is fixed (a car of given mass must be brought from its speed to rest). The only way to **reduce the average force** on the occupants is therefore to **increase the impact time** Δt. This single idea underlies most modern safety engineering:

- **Airbags**: the driver's head sinks into the inflating bag, so it stops over perhaps 0.1 s instead of 0.01 s against the steering wheel – the average force is divided by about ten.
- **Crumple zones**: the front and rear of a vehicle are designed to deform progressively, lengthening the collision time while the passenger cell stays rigid.
- **Seatbelts**: they stretch slightly and spread the force over the strong parts of the body (chest, pelvis), extending the stopping time.
- **Bending the knees on landing**: a paratrooper or basketball player who lands with bent legs extends the stopping time of his body's momentum, sparing his joints.
- **Padded goalposts and crash mats**: the padding compresses, increasing Δt and reducing F_{avg}.



The airbag principle: both force–time curves enclose the same area (same change of momentum), but the airbag spreads the impact over a longer time, so the peak and average forces are much smaller.

WORKED EXAMPLE 3 **Why the goalkeeper pulls his hands back**

A 0.43 kg football arrives at a goalkeeper's hands at 25 m s^{-1} and is stopped. (a) Find the change of momentum. (b) Find the average force on the hands if the ball is stopped in 0.02 s (rigid hands) and in 0.15 s (hands drawn back).

SOLUTION

Solution. Take the ball's initial direction as positive: $u = 25$ m s^{-1}, $v = 0$.

(a) $\Delta p = m(v - u) = 0.43(0 - 25) = -10.75$ kg m s^{-1}; **magnitude** 10.75 kg m s^{-1}.

(b) Rigid hands: $F_{\text{avg}} = \dfrac{10.75}{0.02} \approx 538$ N – painful!

Hands drawn back: $F_{\text{avg}} = \dfrac{10.75}{0.15} \approx 72$ N – a mere sting.

Multiplying the impact time by 7.5 divides the force by 7.5: this is exactly why good goalkeepers "give" with the ball.

WARNING **Impulse vs. force: do not confuse them**

Impulse (N s) measures the total effect of a force over time and equals the change of momentum; force (N) is the rate of change of momentum. Writing $J = F\Delta t$ is only valid for a constant force (or when F is the average force). For a varying force you must use the area under the F–t graph or the integral $\int F\,dt$.

KEY POINTS **Key points of this section**

- Linear momentum: $\vec{p} = m\vec{v}$, a vector in the direction of motion; unit kg m s^{-1} ≡ N s.
- Total momentum of a system = vector sum of individual momenta; in 1-D, use signs.
- Newton's second law, general form: $\vec{F} = \dfrac{d\vec{p}}{dt}$; it reduces to $\vec{F} = m\vec{a}$ only when mass is constant.
- Impulse: $\vec{J} = \vec{F}\Delta t$ (constant force) or $\vec{J} = \int \vec{F}\,dt$ (variable force); equals change of momentum $\Delta\vec{p}$.
- Impulse = area under F–t graph.
- To reduce impact force, increase impact time (airbags, crumple zones, seatbelts, bending knees).

2 CONSERVATION OF LINEAR MOMENTUM

In the previous section we established that the momentum $\vec{p} = m\vec{v}$ of a body changes only when a net external impulse acts on it, and that $\vec{F}_{\text{net}} = \dfrac{d\vec{p}}{dt}$. We now ask a deeper question: what happens when two or more bodies interact with each other? The answer is one of the most powerful principles in all of physics — the **principle of conservation of linear momentum**. It allows us to predict the outcome of collisions, explosions and recoils without knowing any detail of the forces involved.

2.1 Internal and External Forces: the Idea of a System

EXPERIMENT **Two skaters push each other**

Imagine two students on roller skates, standing face to face on a smooth floor. Student A pushes student B. Both students move — B moves forward and A moves backward. If we look at each student separately, each one's momentum has changed, because each experienced a force from the other. But if we look at the pair together, something remarkable happens: the total momentum of the pair is still zero, exactly as it was before the push. The push was an internal affair of the pair.

To make this precise we must first say what we mean by a **system**.

DEFINITION **System, internal and external forces**

A **system** is any collection of bodies that we choose to consider together. Forces acting on the bodies of the system are of two kinds:

- **Internal forces**: forces exerted between bodies of the system (e.g. the contact forces between two colliding trolleys, the explosive forces between fragments of a shell).
- **External forces**: forces exerted on bodies of the system by agents outside the system (e.g. weight, normal reaction of the ground, friction from the road, air resistance).

DEFINITION **Isolated (closed) system**

An **isolated system** (or closed system) is a system on which the resultant external force is zero:

$$\sum \vec{F}_{\text{ext}} = \vec{0}.$$

This happens either because no external forces act at all, or because the external forces cancel one another (e.g. weight balanced by the normal reaction on a smooth horizontal table).

WARNING **Choosing the system is your decision**

The same interaction can be isolated or not, depending on what you include in the system. A single billiard ball struck by another is not isolated (it feels the contact force from outside). The pair of billiard balls on a smooth table is isolated, because the contact forces are now internal and the external forces (weight, reaction) cancel.

2.2 Statement of the Principle

PROPERTY **Principle of conservation of linear momentum**

If the resultant external force on a system is zero (isolated system), then the total linear momentum of the system remains constant:

$$\vec{P}_{\text{total}} = \vec{p}_1 + \vec{p}_2 + \cdots + \vec{p}_n = \text{constant},$$

that is,

$$\boxed{\vec{P}_{\text{before}} = \vec{P}_{\text{after}}}$$

for any two instants before and after an internal interaction. This is a vector statement: it holds component by component along each independent direction. The derivation from Newton's laws is examinable.

2.3 Derivation from Newton's Second and Third Laws

Consider two bodies A and B, of masses m_1 and m_2, forming an isolated system. During their interaction, body A exerts a force $\vec{F}_{A\text{ on }B}$ on B, and B exerts $\vec{F}_{B\text{ on }A}$ on A.

Step 1 — Newton's third law. The interaction forces are equal and opposite at every instant:

$$\vec{F}_{B\text{ on }A} = -\vec{F}_{A\text{ on }B}.$$

Step 2 — Newton's second law in momentum form. Since the system is isolated, these internal forces are the only forces acting on each body, so

$$\vec{F}_{B\text{ on }A} = \frac{d\vec{p}_1}{dt}, \qquad \vec{F}_{A\text{ on }B} = \frac{d\vec{p}_2}{dt}.$$

Step 3 — Add. Adding the two equations and using Step 1:

$$\frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} = \vec{F}_{B\text{ on }A} + \vec{F}_{A\text{ on }B} = \vec{0}.$$

Hence

$$\frac{d}{dt}\left(\vec{p}_1 + \vec{p}_2\right) = \vec{0} \quad \Longrightarrow \quad \vec{p}_1 + \vec{p}_2 = \text{constant}. \qquad \blacksquare$$

The argument extends immediately to n bodies, because internal forces always occur in Newton's-third-law pairs whose vector sum is zero. Notice what the proof reveals: internal forces redistribute momentum between the bodies, but can never change the total.

KEY POINTS **The vector nature of conservation**

Conservation of momentum is a vector law. In two or three dimensions it gives one scalar

equation per axis:

$$\sum m_i u_{ix} = \sum m_i v_{ix}, \qquad \sum m_i u_{iy} = \sum m_i v_{iy}.$$

Momentum along each independent direction is conserved separately. A body can gain momentum in the x-direction while the total y-momentum stays zero.

2.4 A Working Routine

METHOD **Conservation-of-momentum routine**

1. Define the system: decide which bodies you will consider together.
2. Check isolation: confirm that the resultant external force is zero, or that external impulses are negligible during the short interaction.
3. Draw before/after diagrams: sketch the situation just before and just after the interaction, labelling all velocities.
4. Choose a positive direction (one per axis) and stick to it; velocities in the opposite direction carry a minus sign.
5. Write the vector equation $\vec{P}_{\text{before}} = \vec{P}_{\text{after}}$, i.e. $\sum m_i \vec{u}_i = \sum m_i \vec{v}_i$, and solve component by component.
6. Interpret the sign of your answer: a negative velocity means motion opposite to the chosen positive direction.

2.5 Applications in One Dimension

Two bodies pushing apart. Two trolleys of masses m_1 and m_2, initially at rest and touching, are pushed apart by a spring between them. The initial total momentum is zero, so

$$0 = m_1 v_1 + m_2 v_2 \quad \Longrightarrow \quad \frac{v_1}{v_2} = -\frac{m_2}{m_1}.$$

The lighter trolley moves faster, in the opposite direction. The momenta are equal in magnitude and opposite in direction.

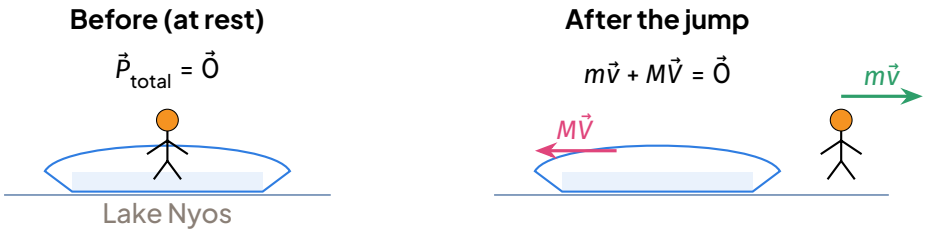
Man jumping from a canoe. A fisherman of mass m stands on a stationary canoe of mass M on calm water (resistance negligible). He jumps horizontally off the stern with velocity $\vec{v}$ relative to the ground. The system {man + canoe} is isolated horizontally, with zero initial momentum:

$$0 = m\vec{v} + M\vec{V} \quad \Longrightarrow \quad \vec{V} = -\frac{m}{M}\vec{v}.$$

The canoe glides backwards — this is why stepping carelessly out of a dugout canoe on Lake Nyos can leave you in the water while the canoe drifts away!

WARNING **Velocities are measured relative to the ground**

In all conservation-of-momentum equations, every velocity must be expressed in the same reference frame, normally the ground (or the water at rest). If a problem gives the man's velocity relative to the canoe, you must first convert it to a ground-frame velocity before substituting.



Man jumping from a stationary canoe: the forward momentum of the man is balanced by the backward momentum of the canoe, so the total momentum remains zero.

WORKED EXAMPLE 4 **Shunting railway wagons**

A railway wagon of mass 12 000 kg moving at 4.0 m s^{-1} on a straight level track collides with a stationary wagon of mass 8 000 kg, and the two couple together. Find their common velocity just after coupling.

SOLUTION

System: the two wagons. Isolation: during the brief coupling, external impulses (friction) are negligible; weight cancels the reaction.
Take the initial direction of motion as positive.

$$P_{\text{before}} = 12000 \times 4.0 + 8000 \times 0 = 48000 \text{ kg m s}^{-1},$$
$$P_{\text{after}} = (12000 + 8000)\,v = 20000\,v.$$

Conservation of momentum:

$$20000\,v = 48000 \quad \Longrightarrow \quad v = 2.4 \text{ m s}^{-1}.$$

The coupled wagons move on at 2.4 m s^{-1} in the original direction. (Note: kinetic energy is lost here — the coupling is inelastic — yet momentum is exactly conserved.)

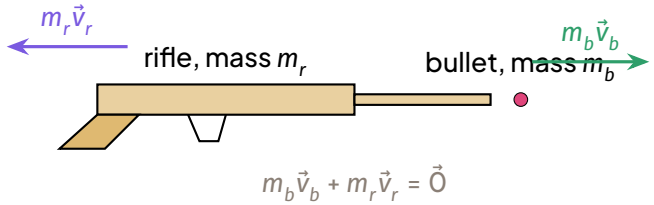
2.6 Recoil of a Gun

When a rifle fires, the expanding gases push the bullet forward and the rifle backward with equal

and opposite forces (Newton's third law). The system {rifle + bullet} is isolated during the very short firing, and its total momentum, initially zero, stays zero:

$$m_b\vec{v}_b + m_r\vec{v}_r = \vec{0} \quad\Longrightarrow\quad \vec{v}_r = -\frac{m_b}{m_r}\vec{v}_b.$$

The **recoil velocity** of the rifle is small because its mass is large compared with the bullet's — but the momenta of bullet and rifle are equal in magnitude.



Recoil of a rifle: after firing, the forward momentum of the bullet and the backward momentum of the rifle are equal in magnitude and opposite in direction.

WORKED EXAMPLE 5 **Computing the recoil velocity**

A rifle of mass 4.0 kg fires a bullet of mass 10 g with a muzzle velocity of 400 m s^{-1}. The rifle is held loosely. Calculate the recoil velocity of the rifle.

SOLUTION

The system {rifle + bullet} is initially at rest: $P_{\text{before}} = 0$. Taking the bullet's direction as positive, with $m_b = 0.010$ kg:

$$0 = m_b v_b + m_r v_r \quad\Longrightarrow\quad v_r = -\frac{m_b v_b}{m_r} = -\frac{0.010 \times 400}{4.0} = -1.0\ \text{m s}^{-1}.$$

The rifle recoils at 1.0 m s^{-1} opposite to the bullet's motion. The negative sign indicates the direction of recoil. (In practice a firm shoulder adds an external impulse and reduces the free recoil.)

2.7 Condition of Validity: Negligible External Impulse

Strictly, conservation requires $\sum \vec{F}_{\text{ext}} = \vec{0}$. In real collisions there is often friction or gravity acting. Why can we still apply the principle?

The change in total momentum equals the impulse of external forces:

$$\Delta\vec{P} = \vec{J}_{\text{ext}} = \sum \vec{F}_{\text{ext}}\,\Delta t.$$

During a collision or explosion, Δt is extremely short (milliseconds), while the internal forces are enormous. The external impulse $\vec{F}_{\text{ext}}\Delta t$ is then negligible compared with the momenta involved,

so

$$\vec{P}_{\text{just before}} \approx \vec{P}_{\text{just after}}.$$

WARNING **Apply conservation only across the short interaction**

Conservation of momentum is applied between instants immediately before and immediately after the collision or explosion. Do not extend it over long time intervals during which friction, gravity or other external forces have time to act significantly.

2.8 Conservation of Momentum versus Conservation of Energy

A frequent source of confusion at Advanced Level is the difference between these two great conservation laws.

	Linear momentum	Kinetic energy
Nature	Vector ($m\vec{v}$)	Scalar ($\frac{1}{2}mv^2$)
Conserved in isolated system?	**Always** (Newton's third law)	Only if the collision is **elastic**
In inelastic collisions / explosions	Conserved	Converted to heat, sound, deformation (or released by explosives)

In the shunting example above, the kinetic energy before coupling was $\frac{1}{2}(12000)(4.0)^2 = 96\,000$ J, while afterwards it is $\frac{1}{2}(20000)(2.4)^2 = 57\,600$ J: some 38 400 J was dissipated in the coupling mechanism — yet momentum was conserved exactly.

WARNING **Momentum is conserved for the SYSTEM, not for each body**

Each colliding body generally changes its momentum — that is the whole point of the interaction. It is the vector sum of the momenta of all bodies in the isolated system that stays constant. Writing $m_1u_1 = m_1v_1$ for a single body is a serious error.

WARNING **Never assume kinetic energy is conserved**

Unless the collision is stated (or shown) to be elastic, do not write $\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$. Momentum conservation alone, applied as a vector equation, is the safe universal tool; energy conservation is a bonus valid only for elastic interactions.

DID YOU KNOW? **Why conservation of momentum is so fundamental**

Newton's laws are an approximation valid at ordinary speeds, but the conservation of linear momentum survives everywhere: in relativistic physics, in quantum mechanics, and in the collisions of galaxies. Physicists regard it as a consequence of a deep symmetry of nature — the fact that the laws of physics are the same everywhere in space (homogeneity of space). When you solve a GCE collision problem, you are using one of the deepest principles known to science!

EXERCISE 1 **Practice**

1. A man of mass 70 kg stands at rest on a stationary canoe of mass 90 kg. He jumps off horizontally at 2.5 m s^{-1} relative to the ground. Find the velocity of the canoe just after he leaves it. (Water resistance negligible.)
2. A gun of mass 5.0 kg, free to move, fires a 20 g shell at 250 m s^{-1}. Find the recoil speed of the gun and the ratio of the kinetic energies of shell and gun. Comment.
3. A trolley of mass 2.0 kg moving at 3.0 m s^{-1} catches up with a trolley of mass 1.0 kg moving at 1.0 m s^{-1} in the same direction; they stick together. Find their common velocity and the kinetic energy lost.

SOLUTION **Exercise 1**

System {man + canoe}, isolated horizontally, initial momentum zero. Taking the man's jump direction as positive:

$$0 = 70(2.5) + 90V \quad\Longrightarrow\quad V = -\frac{175}{90} \approx -1.94\ \text{m s}^{-1}.$$

The canoe moves backwards at about 1.9 m s^{-1}.

SOLUTION **Exercise 2**

$0 = (0.020)(250) + 5.0\,v_r \Rightarrow v_r = -1.0$ m s^{-1}; recoil speed 1.0 m s^{-1}. Kinetic energies: shell = $\tfrac{1}{2}(0.020)(250)^2 = 625$ J; gun = $\tfrac{1}{2}(5.0)(1.0)^2 = 2.5$ J. Ratio 250 : 1. Although the momenta are equal, the light shell carries almost all the kinetic energy — equal momenta do not mean equal energies.

SOLUTION **Exercise 3**

Momentum: $2.0(3.0) + 1.0(1.0) = 3.0\,v \Rightarrow v = \dfrac{7.0}{3.0} \approx 2.33\ \text{m s}^{-1}$. KE before $= \frac{1}{2}(2.0)(9) + \frac{1}{2}(1.0)(1) = 9.5$ J; KE after $= \frac{1}{2}(3.0)\left(\frac{7}{3}\right)^2 \approx 8.17$ J. Energy lost ≈ 1.33 J, dissipated as heat, sound and deformation in the coupling.

3 COLLISIONS IN ONE DIMENSION: ELASTIC AND INELASTIC

In the previous section we established the **Principle of Conservation of Linear Momentum**: whenever two bodies interact and no external force acts on the system, the total momentum before the interaction equals the total momentum after it. We now put this principle to work on the most important family of interactions in mechanics: **collisions**. A collision is a brief, intense interaction between two bodies during which they exert large forces on each other for a short time. Think of two trolleys meeting on a linear air track in the school laboratory, of two lorries crashing on the Douala–Yaoundé highway, or of a bullet striking a wooden block. In every case momentum is conserved; what distinguishes one collision from another is what happens to the **kinetic energy**.

3.1 Classifying Collisions

DEFINITION **Elastic, inelastic and perfectly inelastic collisions**

A **collision** is an interaction between two bodies in which they exert forces on each other over a short time interval.

- A collision is **elastic** if kinetic energy is conserved: the total kinetic energy before impact equals the total kinetic energy after impact.
- A collision is **inelastic** if some kinetic energy is lost (converted into heat, sound and permanent deformation).
- A collision is **perfectly inelastic** (or coalescing) if the two bodies stick together after impact and move with a common velocity. This is the case of maximum kinetic energy loss consistent with momentum conservation.

In all collisions, total linear momentum is conserved (provided no external impulse acts).

WARNING **Momentum yes, kinetic energy not always**

The single most common error at GCE is to write "kinetic energy before = kinetic energy after" for every collision. Kinetic energy is conserved only in elastic collisions. If the bodies stick together, or if the question mentions heat, sound or deformation, kinetic energy is not conserved. Momentum, however, is conserved in every collision.

3.2 Elastic Collisions in One Dimension

Consider two particles of masses m_1 and m_2 moving along the same straight line with velocities u_1 and u_2 ($u_1 > u_2$, so that they collide). After the elastic collision their velocities are v_1 and v_2. Two conservation laws apply:

$$\begin{aligned} \text{Momentum:} \quad & m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 \\ \text{Kinetic energy:} \quad & \tfrac{1}{2}m_1u_1^2 + \tfrac{1}{2}m_2u_2^2 = \tfrac{1}{2}m_1v_1^2 + \tfrac{1}{2}m_2v_2^2 \end{aligned}$$

These are two simultaneous equations in the two unknowns v_1 and v_2. The energy equation contains squares, which makes direct solution heavy; fortunately there is a beautiful shortcut.

PROPERTY **Speed of approach equals speed of separation**

In a one-dimensional elastic collision,

$$u_1 - u_2 = v_2 - v_1,$$

that is, the **speed of approach** before impact equals the **speed of separation** after impact.

Proof (examinable). Rearrange the momentum equation by grouping terms in each mass:

$$m_1(u_1 - v_1) = m_2(v_2 - u_2). \tag{1}$$

Rearrange the energy equation in the same way, cancelling the factors $\tfrac{1}{2}$:

$$m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2).$$

Factorising each difference of squares:

$$m_1(u_1 - v_1)(u_1 + v_1) = m_2(v_2 - u_2)(v_2 + u_2). \tag{2}$$

Dividing (2) by (1) (both sides of (1) are non-zero during a genuine collision) gives

$$u_1 + v_1 = v_2 + u_2 \quad \Longrightarrow \quad u_1 - u_2 = v_2 - v_1. \qquad \blacksquare$$

This is exactly **Newton's experimental law of restitution** with coefficient of restitution $e = 1$.

METHOD **Strategy for elastic collisions**

1. Draw a clear before/after diagram; fix a positive direction and write every velocity with its sign.
2. Write the momentum equation: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.
3. Write the restitution equation $u_1 - u_2 = v_2 - v_1$ (much easier than the energy equation, and equivalent to it for elastic impacts).
4. Solve the two linear equations simultaneously for v_1 and v_2.
5. Check: verify that kinetic energy is indeed conserved and that the answer is physically sensible (the front body must end up faster than the rear body, otherwise they have passed through each other!).

EXEMPLE **Special case 1: equal masses exchange velocities**

Set $m_1 = m_2 = m$. The momentum equation gives $u_1 + u_2 = v_1 + v_2$, and restitution gives $u_1 - u_2 = v_2 - v_1$. Adding and subtracting these yields

$$v_1 = u_2, \qquad v_2 = u_1.$$

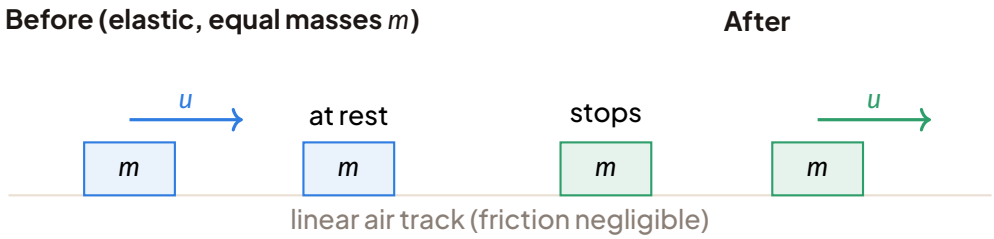
The two bodies simply swap velocities. If the target was at rest ($u_2 = 0$), the incoming body stops dead and the target moves off with the incoming velocity. This is exactly what you see when a moving trolley hits an identical stationary trolley on an air track.

EXEMPLE **Special case 2: light body bouncing off a massive fixed body**

Let $m_2 \to \infty$ with $u_2 = 0$ (a wall, or the Earth). Solving the general equations (or taking limits) gives

$$v_1 \approx -u_1, \qquad v_2 \approx 0.$$

The light body rebounds with (nearly) the same speed, in the opposite direction, while the massive body barely moves. A tennis ball dropped onto a hard floor bouncing back to nearly the same height is a familiar example.



Elastic collision of two equal-mass trolleys: the incoming trolley stops and the stationary one moves off with velocity u — the trolleys exchange velocities.

WORKED EXAMPLE 6 **Head-on elastic collision**

A car of mass 1200 kg travelling at 20 m s^{-1} collides head-on elastically (an idealisation!) with a car of mass 800 kg travelling at 10 m s^{-1} in the opposite direction. Find their velocities after the collision.

SOLUTION

Solution. Take the direction of the 1200 kg car as positive. Then

$$m_1 = 1200,\ u_1 = 20; \qquad m_2 = 800,\ u_2 = -10.$$

Momentum:

$$1200(20) + 800(-10) = 1200v_1 + 800v_2 \implies 16000 = 1200v_1 + 800v_2,$$

i.e. $3v_1 + 2v_2 = 40$. (i)
Restitution (elastic): $u_1 - u_2 = v_2 - v_1$:

$$20 - (-10) = v_2 - v_1 \implies v_2 - v_1 = 30. \quad (ii)$$

From (ii), $v_2 = v_1 + 30$; substituting in (i): $3v_1 + 2(v_1 + 30) = 40$, so $5v_1 = -20$, giving

$$v_1 = -4\ \text{m s}^{-1}, \qquad v_2 = 26\ \text{m s}^{-1}.$$

The heavy car rebounds at 4 m s^{-1}; the light car bounces back at 26 m s^{-1}.
Check (energy): $\text{KE}_{\text{before}} = \frac{1}{2}(1200)(400) + \frac{1}{2}(800)(100) = 280000$ J; $\text{KE}_{\text{after}} = \frac{1}{2}(1200)(16) + \frac{1}{2}(800)(676) = 9600 + 270400 = 280000$ J. ✓

3.3 Perfectly Inelastic Collisions: Bodies Stick Together

When two vehicles crumple and lock together in a crash, or when a bullet embeds itself in a block of wood, the bodies move afterwards with a **common velocity** v. Only one unknown re-

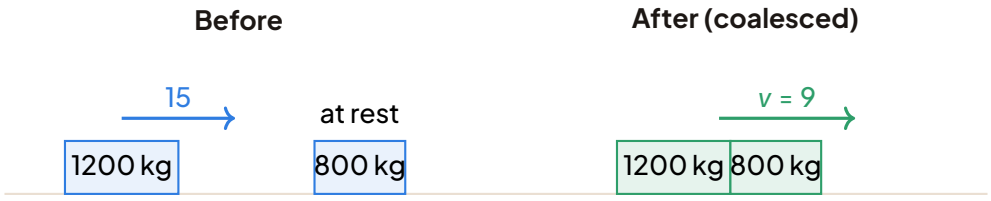
mains, so momentum conservation alone suffices:

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v \quad \Longrightarrow \quad v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}.$$

Kinetic energy is not conserved: part of it is converted into heat, sound and the work of permanently deforming the bodies.

METHOD **Computing the kinetic energy lost**

1. Use momentum conservation to find the final velocity (or velocities).
2. Compute $\mathrm{KE}_{\text{before}} = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2$ and $\mathrm{KE}_{\text{after}}$.
3. Energy lost: $\Delta\mathrm{KE} = \mathrm{KE}_{\text{before}} - \mathrm{KE}_{\text{after}}$.
4. Express the loss as a fraction or percentage of the initial KE: $\dfrac{\Delta\mathrm{KE}}{\mathrm{KE}_{\text{before}}} \times 100\%$.
5. State where the energy goes: heat, sound, permanent deformation.



Perfectly inelastic crash: a 1200 kg vehicle at 15 $\mathrm{m\,s^{-1}}$ strikes a stationary 800 kg vehicle; they lock together. Momentum gives $v = \dfrac{1200 \times 15}{2000} = 9\ \mathrm{m\,s^{-1}}$.

WORKED EXAMPLE 7 **Kinetic energy lost in a coalescing crash**

Using the collision of the figure above: a 1200 kg car moving at 15 $\mathrm{m\,s^{-1}}$ hits a stationary 800 kg car and the two lock together. Find (a) the common velocity, (b) the kinetic energy lost, (c) the percentage of the initial KE lost.

SOLUTION

(a) Momentum: $1200(15) + 800(0) = 2000\,v \Rightarrow v = 9\ \mathrm{m\,s^{-1}}$.
(b) $\mathrm{KE}_{\text{before}} = \frac{1}{2}(1200)(15^2) = 135000$ J. $\mathrm{KE}_{\text{after}} = \frac{1}{2}(2000)(9^2) = 81000$ J. $\Delta\mathrm{KE} = 135000 - 81000 = 54000$ J, converted into heat, sound and deformation of the bodywork.
(c) Percentage lost $= \dfrac{54000}{135000} \times 100\% = 40\%$.

WORKED EXAMPLE 8 **The ballistic pendulum idea**

A bullet of mass 20 g travelling horizontally at 350 m s^{-1} embeds itself in a wooden block of mass 1.98 kg resting on a smooth horizontal table. Find (a) the velocity of block and bullet just after impact, (b) the fraction of the bullet's kinetic energy that is lost.

SOLUTION

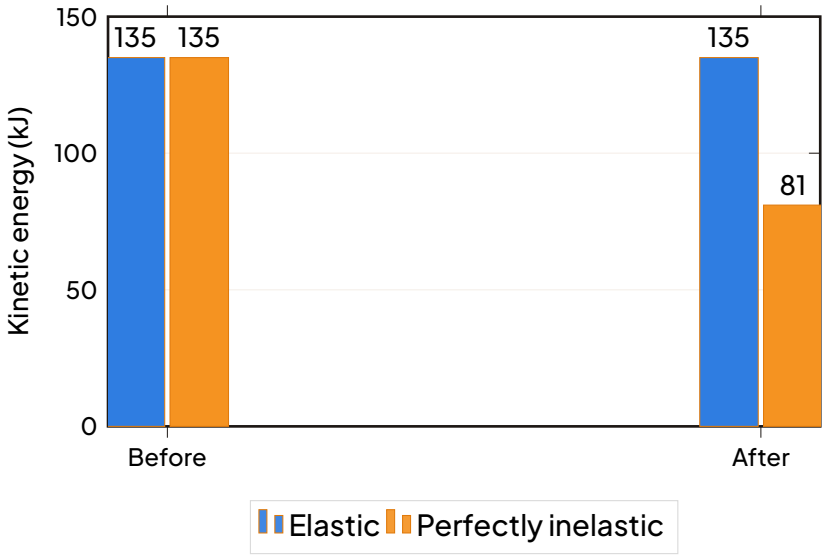
(a) Momentum (bullet + block, perfectly inelastic):

$$0.020(350) = (0.020 + 1.98)\,v \implies v = \frac{7.0}{2.00} = 3.5\ \text{m s}^{-1}.$$

(b) $\text{KE}_{\text{before}} = \tfrac{1}{2}(0.020)(350^2) = 1225$ J; $\text{KE}_{\text{after}} = \tfrac{1}{2}(2.00)(3.5^2) = 12.25$ J. Fraction lost $= \dfrac{1225 - 12.25}{1225} = 0.99$: 99% of the kinetic energy is dissipated as heat, sound and the work of tearing the wood. (In a real ballistic pendulum the block then swings upward; the common velocity found here is the starting point for that energy calculation.)

3.4 Comparing Elastic and Perfectly Inelastic Collisions

The bar chart below compares the kinetic energy before and after impact for the two archetypal collisions studied above, each starting with the same 1200 kg vehicle at 15 m s^{-1} striking a stationary 800 kg body.



Kinetic energy before and after impact. Elastic collision: KE is fully conserved (135 kJ throughout). Perfectly inelastic collision: KE falls from 135 kJ to 81 kJ — 40% is dissipated as heat, sound and deformation.

3.5 Extension: The Coefficient of Restitution

Real collisions lie between the elastic and perfectly inelastic extremes. **Newton's experimental law** states that for a direct impact, the speed of separation is a constant fraction e of the speed of approach:

$$v_2 - v_1 = e\,(u_1 - u_2), \qquad 0 \le e \le 1,$$

where e is the **coefficient of restitution**, a property of the materials in contact. The limiting cases are exactly the ones we have studied: $e = 1$ gives a perfectly **elastic** collision (speed of separation = speed of approach), and $e = 0$ gives a **perfectly inelastic** collision ($v_2 = v_1$, the bodies move together). Together with momentum conservation, this law solves any one-dimensional collision — a very useful tool for GCE problems involving bouncing balls.

WARNING **Exam traps**

- Never write $\mathrm{KE}_{\text{before}} = \mathrm{KE}_{\text{after}}$ when the bodies stick together — use momentum only.
- Velocities are signed: in a head-on collision one of the u's must be negative.
- In the restitution law, approach uses initial velocities and separation uses final velocities, in consistent order: $v_2 - v_1 = e(u_1 - u_2)$.
- After the collision the rear body cannot be faster than the front body — if your algebra says otherwise, recheck your signs.

KEY POINTS **Key points**

- Momentum is conserved in every collision; kinetic energy only in elastic ones.
- Elastic 1-D collision: solve $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ together with $u_1 - u_2 = v_2 - v_1$.
- Equal masses exchange velocities; a light body rebounds off a massive fixed body with nearly the same speed.
- Perfectly inelastic: common velocity $v = \dfrac{m_1u_1 + m_2u_2}{m_1 + m_2}$; the lost KE becomes heat, sound and deformation.
- General law: $v_2 - v_1 = e(u_1 - u_2)$, with $e = 1$ elastic, $e = 0$ perfectly inelastic.

EXEMPLE **Exam tip**

At the GCE, always state explicitly which principle you are using and why: "By conservation of linear momentum (no external impulse during the impact)…" and, for elastic collisions, "Since the collision is elastic, kinetic energy is conserved (equivalently, speed

of approach = speed of separation)…". Examiners award method marks for these statements even before any calculation.

EXERCISE 2

A trolley of mass 2.0 kg moving at 6.0 m s^{-1} on a smooth track collides with a stationary trolley of mass 4.0 kg. (a) Assuming the collision is perfectly elastic, find both final velocities. (b) Assuming instead that the trolleys stick together, find their common velocity and the percentage of kinetic energy lost.

SOLUTION **Exercise 1**

(a) Momentum: $2(6) = 2v_1 + 4v_2 \Rightarrow v_1 + 2v_2 = 6$. Restitution: $v_2 - v_1 = 6$. Solving: $v_2 = 4$ m s^{-1}, $v_1 = -2$ m s^{-1} (the light trolley rebounds). (b) Coalescing: $12 = 6v \Rightarrow v = 2$ m s^{-1}. $\mathrm{KE}_{\text{before}} = \tfrac{1}{2}(2)(36) = 36$ J; $\mathrm{KE}_{\text{after}} = \tfrac{1}{2}(6)(4) = 12$ J. Loss $= \dfrac{24}{36} \times 100\% \approx 66.7\%$.

4 COLLISIONS AND MOMENTUM IN TWO DIMENSIONS

In the previous section we studied collisions along a straight line: two bodies approaching each other on the same road, or a ball striking another head-on. Real life, however, is rarely so tidy. Two cars meet at a crossroads in Buea at right angles; a snooker ball clips another and both roll off in different directions; an alpha particle is scattered sideways by a nucleus. In all these situations the velocities before and after the impact do not lie on one line, and the one-dimensional equation $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ is no longer enough. We must treat momentum as what it truly is: a **vector**. This section develops the full vector machinery of two-dimensional collisions, which is one of the most frequently examined topics in the GCE Advanced Level mechanics papers.

4.1 Momentum as a Plane Vector

DEFINITION **Momentum in component form**

The momentum of a body of mass m moving with velocity $\vec{v} = v_x\,\vec{i} + v_y\,\vec{j}$ is the vector

$$\vec{p} = m\vec{v} = mv_x\,\vec{i} + mv_y\,\vec{j} = \begin{pmatrix} mv_x \\ mv_y \end{pmatrix}.$$

Its magnitude is $|\vec{p}| = m\sqrt{v_x^2 + v_y^2} = mv$ and its direction is that of $\vec{v}$.

The key observation is simple: since mass is a scalar, multiplying a velocity vector by m does not change its direction. Everything we know about resolving vectors therefore carries over directly to momentum. If a ball of mass 0.5 kg moves at 10 m s^{-1} at 30° above the horizontal, its momentum has components

$$p_x = 0.5 \times 10 \cos 30^\circ = 2.5\sqrt{3}\,\text{kg m s}^{-1}, \qquad p_y = 0.5 \times 10 \sin 30^\circ = 2.5\,\text{kg m s}^{-1}.$$

PROPERTY **Conservation of momentum along each axis**

In any collision or explosion where no external impulse acts, the total momentum vector is conserved:

$$\sum \vec{p}_{\text{before}} = \sum \vec{p}_{\text{after}}.$$

A vector equation in the plane is equivalent to two independent scalar equations: the total x-component of momentum is conserved, and the total y-component of momentum is conserved, separately:

$$\sum p_x \text{ constant}, \qquad \sum p_y \text{ constant}.$$

This is the single most important idea of the section. Two perpendicular directions cannot "exchange" momentum: whatever momentum the system has along $\vec{i}$ stays along $\vec{i}$, and likewise for $\vec{j}$. In practice this gives us two equations, which is exactly what we need to find two unknowns (typically a speed and an angle).

4.2 The Component Method for 2-D Collisions

METHOD **2-D collision routine**

1. Choose perpendicular axes $\vec{i}$ and $\vec{j}$ (usually along and perpendicular to the initial motion, or along the line of centres).
2. Resolve every velocity, before and after, into components. Draw a clear "before" and "after" sketch with all angles marked.
3. Write conservation of momentum along $\vec{i}$: one scalar equation.
4. Write conservation of momentum along $\vec{j}$: a second scalar equation.
5. Solve the two simultaneous equations for the unknown components.
6. Recombine: speed $= \sqrt{v_x^2 + v_y^2}$, direction $= \tan^{-1}\left|\dfrac{v_y}{v_x}\right|$ measured from the chosen axis.

State the angle clearly.

WORKED EXAMPLE 9 **Two cars at a crossroads**

A car A of mass 1200 kg travelling due east at 20 m s^{-1} collides at a junction with a car B of mass 800 kg travelling due north at 15 m s^{-1}. The cars lock together on impact. Find the velocity of the wreckage immediately after the collision.

SOLUTION

Solution. Take $\vec{i}$ due east and $\vec{j}$ due north. Let the common velocity after impact be $\vec{v} = v_x\vec{i} + v_y\vec{j}$.

Along $\vec{i}$ (only A has eastward momentum before):

$$1200 \times 20 = (1200 + 800)v_x \implies v_x = \frac{24000}{2000} = 12\,\text{m s}^{-1}.$$

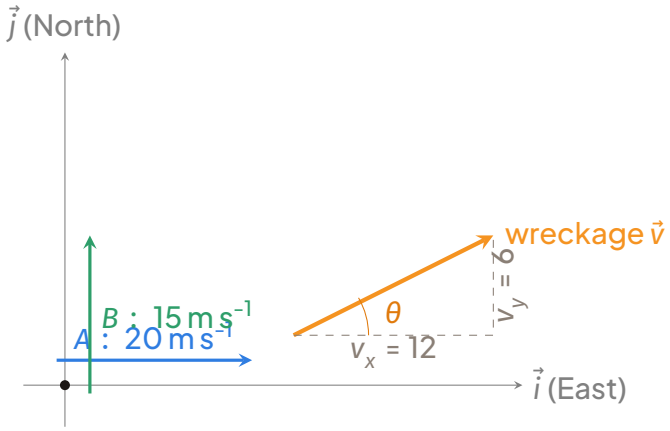
Along $\vec{j}$ (only B has northward momentum before):

$$800 \times 15 = 2000\,v_y \implies v_y = \frac{12000}{2000} = 6\,\text{m s}^{-1}.$$

Speed of the wreckage:

$$v = \sqrt{12^2 + 6^2} = \sqrt{180} = 6\sqrt{5} \approx 13.4\,\text{m s}^{-1}.$$

Direction: $\tan\theta = \dfrac{v_y}{v_x} = \dfrac{6}{12} = 0.5$, so $\theta \approx 26.6^\circ$ north of east, i.e. on a bearing of approximately 063.4°.



Two cars collide at a junction and move off together; the velocity of the wreckage is found from its components.

WARNING **Direction matters**

A body described as "at rest" or "stopped" after a collision may only have lost its momentum along one axis. Always check both components before concluding that a body is stationary. Likewise, never add momenta as scalars when the motions are not parallel: $\vec{p}_1 + \vec{p}_2$ is a vector sum.

4.3 Oblique Collisions of Smooth Spheres

When two smooth spheres collide, the impulse between them acts along the **line of centres** at the instant of impact (because the spheres are smooth, there is no tangential force). This physical fact leads to a beautiful simplification.

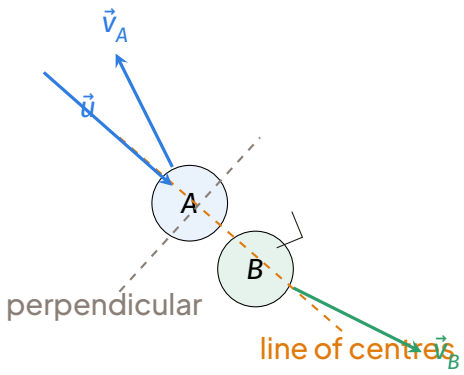
PROPERTY **Oblique impact of smooth spheres**

In a collision between two smooth spheres:

- the component of each sphere's velocity perpendicular to the line of centres is unchanged by the impact;
- along the line of centres, momentum is conserved (and Newton's law of restitution applies) exactly as in a one-dimensional collision.

A particularly clean case occurs when a sphere strikes an identical stationary sphere obliquely. The moving sphere keeps its component of velocity perpendicular to the line of centres; along the line of centres the impact is head-on between equal masses, and for a perfectly elastic impact the spheres simply exchange their line-of-centres velocities. The moving sphere therefore stops dead along the line of centres and moves off purely perpendicular to it, while the

target sphere moves off along the line of centres. The two balls separate at 90° to each other — a result every snooker player knows instinctively.



Sphere A strikes identical stationary sphere B obliquely. The impulse acts along the line of centres; after an elastic impact the spheres move off at right angles.

WORKED EXAMPLE 10 **Snooker-style impact**

A ball of mass 0.16 kg moving at 4 m s^{-1} strikes an identical stationary ball obliquely. At impact, the line of centres makes an angle of 60° with the initial direction of motion. The impact is perfectly elastic. Find the velocities of both balls after impact.

SOLUTION

Solution. Resolve $\vec{u}$ along and perpendicular to the line of centres:

$$u_{\parallel} = 4\cos 60^\circ = 2\ \text{m s}^{-1}, \qquad u_{\perp} = 4\sin 60^\circ = 2\sqrt{3}\ \text{m s}^{-1}.$$

Perpendicular component: unchanged, so the first ball keeps $2\sqrt{3}$ m s^{-1} perpendicular to the line of centres.
Along the line of centres, equal masses, elastic impact, target at rest: the velocities are exchanged, so the first ball's line-of-centres component becomes 0 and the target moves off at 2 m s^{-1} along the line of centres.
Hence the struck ball moves at 2 m s^{-1} along the line of centres (at 60° to the original direction), and the cue ball moves at $2\sqrt{3} \approx 3.46$ m s^{-1} perpendicular to the line of centres (at 30° to the original direction, on the other side). Note 60° + 30° = 90°: the balls separate at right angles, as predicted.

4.4 Momentum Triangles: the Geometric Method

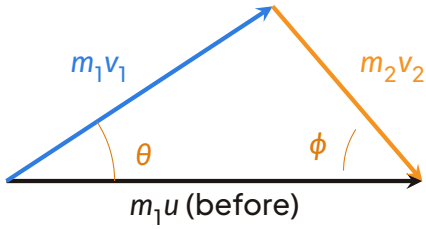
Because momentum is a vector, the equation

$$\vec{p}_1 + \vec{p}_2 = \vec{p}_1' + \vec{p}_2'$$

can be drawn as a closed polygon of vectors — a triangle when only two momenta are involved on each side. Trigonometry (cosine rule and sine rule) then replaces simultaneous equations. This method is often faster and more elegant, especially when one body is initially at rest, so that $\vec{p}_{\text{before}} = \vec{p}_1' + \vec{p}_2'$ forms a single triangle.

METHOD **Momentum–triangle method**

1. Write the vector equation $\sum \vec{p}_{\text{before}} = \sum \vec{p}_{\text{after}}$.
2. Draw the vectors head-to-tail so that the "before" side and the "after" side close the same triangle.
3. Mark all known magnitudes (mu values) and angles.
4. Apply the cosine rule to find an unknown side, and the sine rule to find an unknown angle:
$$a^2 = b^2 + c^2 - 2bc\cos A, \qquad \frac{a}{\sin A} = \frac{b}{\sin B}.$$
5. Convert momentum magnitudes back to velocities by dividing by the relevant mass.



Momentum triangle for a projectile striking a stationary target: $m_1\vec{u} = m_1\vec{v}_1 + m_2\vec{v}_2$. The angles θ and ϕ are the deflections of the two bodies.

WORKED EXAMPLE 11 **Scattering of a particle by a stationary target**

A particle of mass 2 kg moving at 10 m s^{-1} strikes a stationary particle of mass 3 kg. After the collision the 2 kg particle moves at 6 m s^{-1}, deflected through 40° from its original direction. Find the speed and direction of the 3 kg particle.

SOLUTION

Solution by components. Take $\vec{i}$ along the initial direction. Momentum before: $(2 \times 10,\ 0) = (20, 0)$. Momentum of the scattered projectile after: $(2 \times 6\cos 40^\circ,\ 2 \times 6\sin 40^\circ) = (12\cos 40^\circ,\ 12\sin 40^\circ)$.

Let the target move with components (V_x, V_y). Conservation:

$$3V_x = 20 - 12\cos 40^\circ = 20 - 9.193 = 10.807 \implies V_x = 3.602,$$

$$3V_y = 0 - 12\sin 40^\circ = -7.713 \implies V_y = -2.571.$$

Speed: $V = \sqrt{3.602^2 + 2.571^2} = \sqrt{19.58} \approx 4.43\ \text{m s}^{-1}$.
Direction: $\tan\alpha = \dfrac{2.571}{3.602} = 0.714$, so $\alpha \approx 35.5^\circ$ on the opposite side of the original direction from the projectile (the negative sign of V_y).
Check with the triangle: the three momenta 20, 12 and $3V$ close a triangle with 40° between the sides 20 and 12:

$$(3V)^2 = 20^2 + 12^2 - 2(20)(12)\cos 40^\circ = 544 - 367.7 = 176.3,$$

giving $3V = 13.28$, $V \approx 4.43\ \text{m s}^{-1}$ — confirming the component method.

4.5 Is the Collision Elastic? Checking Kinetic Energy in 2–D

Conservation of momentum alone cannot tell us whether a collision is elastic. The test is always the same as in one dimension: compare the total kinetic energy before and after, remembering that kinetic energy is a scalar and uses the full speed:

$$E_k = \tfrac{1}{2}m|\vec{v}|^2 = \tfrac{1}{2}m\left(v_x^2 + v_y^2\right).$$

For the crossroads collision above:

$$E_k^{\text{before}} = \tfrac{1}{2}(1200)(20^2) + \tfrac{1}{2}(800)(15^2) = 240000 + 90000 = 330000\ \text{J},$$

$$E_k^{\text{after}} = \tfrac{1}{2}(2000)(180) = 180000\ \text{J}.$$

Since $180000 < 330000$, kinetic energy is lost: the collision is **inelastic**, as expected for cars that lock together. For the snooker example, $E_k^{\text{before}} = \tfrac{1}{2}(0.16)(16) = 1.28$ J and $E_k^{\text{after}} = \tfrac{1}{2}(0.16)(4) + \tfrac{1}{2}(0.16)(12) = 0.32 + 0.96 = 1.28$ J: energy is conserved, confirming a perfectly elastic impact.

WARNING **Squaring components correctly**

A frequent error is to compute kinetic energy as $\tfrac{1}{2}mv_x^2 + \tfrac{1}{2}mv_y^2$ with the wrong pairing of masses and components, or to forget to recombine components before squaring. Always find $|\vec{v}|^2 = v_x^2 + v_y^2$ for each body first. Also note: momentum can be conserved while kinetic energy is not — never use "energy is conserved" as a substitute for the momentum equations.

EXERCISE 3 **Target at rest, projectile deflected**

A ball of mass 0.5 kg moving at 8 m s^{-1} strikes a stationary ball of mass 1.5 kg. After impact the lighter ball moves at right angles to its original direction at 4 m s^{-1}. Find the velocity of the heavier ball, and determine whether the collision is elastic.

SOLUTION **Exercise 1**

Take $\vec{i}$ along the initial motion, $\vec{j}$ perpendicular. Before: $(0.5\times 8,\ 0) = (4, 0)$. After, light ball: $(0,\ 0.5\times 4) = (0, 2)$. Let the heavy ball have components (V_x, V_y):

$$1.5V_x = 4 - 0 \Rightarrow V_x = \tfrac{8}{3}; \qquad 1.5V_y = 0 - 2 \Rightarrow V_y = -\tfrac{4}{3}.$$

$V = \sqrt{\frac{64}{9} + \frac{16}{9}} = \frac{\sqrt{80}}{3} = \frac{4\sqrt{5}}{3} \approx 2.98$ m s^{-1} at $\tan^{-1}\frac{1}{2} \approx 26.6^\circ$ below the original direction.
Energy: before $= \frac{1}{2}(0.5)(64) = 16$ J; after $= \frac{1}{2}(0.5)(16) + \frac{1}{2}(1.5)\left(\frac{80}{9}\right) = 4 + 6.67 = 10.67$ J < 16 J: inelastic.

KEY POINTS **Key points of the section**

- Momentum is a vector: $\vec{p} = m\vec{v} = (mv_x,\ mv_y)$.
- In 2–D, momentum is conserved independently along each of two perpendicular axes — this gives two scalar equations.
- For smooth spheres, resolve along and perpendicular to the line of centres; the perpendicular components are unchanged.
- The momentum triangle plus the cosine and sine rules offers a fast geometric alternative to components.
- Always state the direction of a final velocity unambiguously, and check kinetic energy to classify the collision as elastic or inelastic.

METHOD **Exam tip: stating the angle of deflection**

GCE questions frequently ask for the angle of deflection or the direction of motion after impact. Never leave the answer as a bare $\tan^{-1}$ value: state precisely what the angle is measured from, e.g. "26.6° north of east", "on a bearing of 063°", or "deflected through 40° from its original path". A correct magnitude with a vague direction often loses a mark.

DID YOU KNOW? **Particle scattering and the atom**

The same mathematics you have just used for snooker balls earned Rutherford's team its place in history: by measuring the angles through which alpha particles were deflected by a thin gold foil and applying conservation of momentum and energy in two dimensions, they deduced that almost all the mass of the atom is concentrated in a tiny nucleus. Vector momentum is truly a tool that reaches from the billiard table to the heart of matter.

5 EXPLOSIONS, ROCKETS AND CENTRE OF MASS

In the previous section we studied collisions, where two moving bodies meet and interact. In this section we turn the situation around: a single body splits apart into fragments. Such events — explosions, nuclear decays, skaters pushing apart — are in a sense **reverse collisions**, and the very same principle of conservation of linear momentum governs them. We then use momentum conservation to explain how rockets accelerate, and we finish with a powerful new concept, the **centre of mass**, which lets us describe the motion of a whole system with a single point.

5.1 Explosions as Reverse Collisions

Observation.

A firework shell is fired vertically and, at the top of its flight, bursts into glowing fragments that fly off in all directions. Before the burst there was one momentum vector; after the burst there are many. Yet no external force acted horizontally during the burst — the explosion is driven by internal forces between the fragments. By Newton's third law these internal forces cancel in pairs, so the total momentum of the system cannot change.

PROPERTY **Momentum conservation in an explosion**

In an explosion, all the forces between the fragments are **internal forces**. Hence the total linear momentum of the system is conserved:

$$\vec{p}_{\text{before}} = \sum_i \vec{p}_{i,\text{after}}.$$

In particular, if the body is initially at rest, the vector sum of the fragments' momenta after the explosion is zero:

$$\sum_i m_i \vec{v}_i = \vec{0}.$$

WARNING **Kinetic energy is NOT conserved**

An explosion is the opposite of an elastic collision. Chemical (or nuclear, or stored elastic) energy is converted into kinetic energy, so the total kinetic energy increases:

$$E_{k,\text{after}} = E_{k,\text{before}} + E_{\text{released}}.$$

Never apply the elastic-collision condition (conservation of kinetic energy, or relative velocity reversed) to an explosion. The only conservation laws you may use are momentum conservation and the energy balance above.

METHOD **Solving explosion problems**

1. Draw a "before" and an "after" diagram; choose axes.
2. Write the total momentum before the explosion (zero if the body was stationary).
3. Write the sum of the fragment momenta after the explosion, with unknown velocities as vectors.
4. Equate the two, component by component, and solve.
5. If the energy released is required, compute $E_{\text{released}} = E_{k,\text{after}} - E_{k,\text{before}}$.

WORKED EXAMPLE 12 **A stationary shell splits into two fragments**

A shell of mass 6 kg, initially at rest, explodes into two fragments of masses 4 kg and 2 kg. The 4 kg fragment moves off at 15 m s^{-1}. Find (a) the velocity of the other fragment, (b) the kinetic energy released by the explosion.

SOLUTION

(a) Initial momentum is zero, so

$$m_1\vec{v}_1 + m_2\vec{v}_2 = \vec{0} \quad\Longrightarrow\quad \vec{v}_2 = -\frac{m_1}{m_2}\vec{v}_1 = -\frac{4}{2}(15) = -30\ \text{m s}^{-1}.$$

The 2 kg fragment moves at 30 m s^{-1} in the opposite direction.
(b) Kinetic energy before: 0. After:

$$E_k = \tfrac{1}{2}(4)(15)^2 + \tfrac{1}{2}(2)(30)^2 = 450 + 900 = 1350\ \text{J}.$$

The explosion released 1350 J of chemical energy as kinetic energy.

WORKED EXAMPLE 13 **A shell exploding into three fragments**

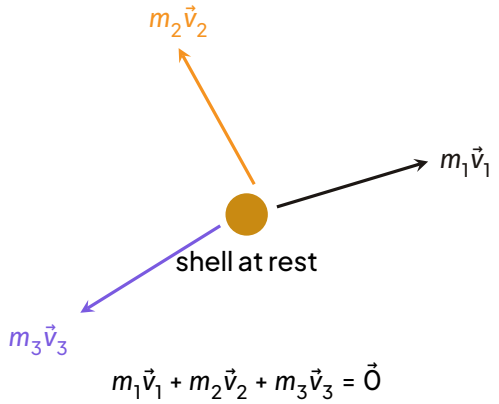
A stationary shell explodes into three fragments A, B, C of masses 1 kg, 2 kg and 3 kg. Fragment A moves at 12 m s^{-1} due east and B moves at 9 m s^{-1} due north. Find the velocity of C.

SOLUTION

Take $\vec{i}$ east, $\vec{j}$ north. Conservation of momentum with zero initial momentum:

$$1(12\vec{i}) + 2(9\vec{j}) + 3\vec{v}_C = \vec{0} \quad \Longrightarrow \quad \vec{v}_C = -4\vec{i} - 6\vec{j}\ \text{m s}^{-1}.$$

Hence $|\vec{v}_C| = \sqrt{4^2 + 6^2} = \sqrt{52} \approx 7.2$ m s^{-1}, directed south-west, at an angle $\tan^{-1}(6/4) \approx 56.3^\circ$ south of west.



A stationary shell explodes into three fragments; the momentum vectors sum to zero.

EXEMPLE **Everyday explosions**

- Skaters pushing apart: two skaters at rest on ice push each other. They glide off in opposite directions with momenta equal in magnitude: $m_1v_1 = m_2v_2$. The lighter skater moves faster.
- Nuclear decay: a stationary nucleus that emits an α-particle recoils in the opposite direction so that total momentum stays zero — this is exactly how the recoil of a gun works too.
- Gun recoil: when a bullet is fired forward, the gun recoils backward. The system (gun + bullet) has zero initial momentum, so $m_{\text{bullet}}v_{\text{bullet}} + m_{\text{gun}}v_{\text{gun}} = 0$.

5.2 Rocket Motion

Observation.

Release an inflated balloon and it darts forward as air rushes out backwards. Nothing pushes the balloon from outside: the expelled air carries momentum backwards, so the balloon must carry equal momentum forwards. A rocket works exactly the same way, burning fuel and ejecting hot exhaust gas at high speed.

PROPERTY **Rocket propulsion by momentum conservation**

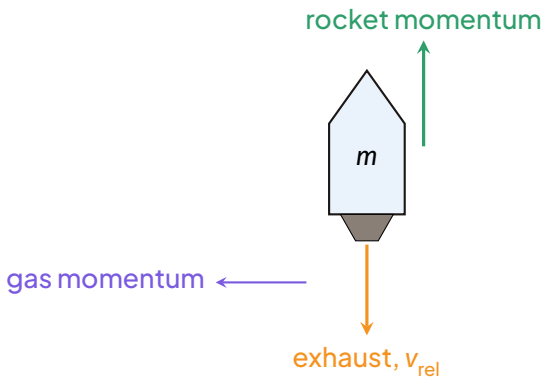
A rocket ejects exhaust gas backwards at speed v_{rel} relative to the rocket. The gas carries momentum backwards; by conservation of momentum the rocket gains momentum forwards. There is no need for anything external to "push against" — a rocket works best in the vacuum of space.

Consider a short time interval Δt during which the rocket ejects a mass Δm of gas at relative speed v_{rel}. The gas gains backward momentum $\Delta m\, v_{\text{rel}}$, so the rocket gains forward momentum $\Delta m\, v_{\text{rel}}$. The rate of change of the rocket's momentum — that is, the **thrust** — is therefore:

PROPERTY **Thrust of a rocket**

$$F = v_{\text{rel}} \frac{dm}{dt},$$

where $\frac{dm}{dt}$ is the rate at which fuel mass is ejected and v_{rel} is the exhaust speed relative to the rocket. The rocket is a **variable-mass** system: as fuel burns, the remaining mass decreases, so the same thrust produces an ever-increasing acceleration $a = F/m(t)$.



A rocket ejects exhaust backwards; momentum conservation drives the rocket forwards.

WORKED EXAMPLE 14 **Computing a thrust**

A rocket engine ejects gas at $2500\ \text{m s}^{-1}$ relative to the rocket at a rate of $80\ \text{kg s}^{-1}$. Find the thrust.

SOLUTION

$$F = v_{\text{rel}}\frac{dm}{dt} = 2500 \times 80 = 2.0e5\ \text{N}.$$

The engine produces a thrust of 200 kN.

EXEMPLE **Multi-stage rockets**

The thrust equation shows why engineers build **multi-stage rockets**. Most of a rocket's initial mass is fuel and the tanks and engines that hold it. Once a stage's fuel is spent, its empty structure is dead weight: carrying it further would waste thrust. By jettisoning the empty stage, the remaining rocket is much lighter, so the next stage's thrust produces a far greater acceleration. Real launch vehicles — such as the Ariane rockets launched from Kourou, or the SpaceX Falcon 9 — use two or three stages for exactly this reason.

DID YOU KNOW? **Beyond the syllabus: the Tsiolkovsky rocket equation**

If you integrate the momentum balance as the rocket's mass falls from its initial value m_0 to a final value m, you obtain the famous **rocket equation** of Konstantin Tsiolkovsky:

$$v = u + v_{\text{rel}}\ln\!\left(\frac{m_0}{m}\right),$$

where u is the initial speed. The logarithm explains why reaching orbit is so hard: each extra bit of speed requires a multiplicative increase in fuel. This derivation is beyond the GCE syllabus, but it is a classic in olympiad problems.

5.3 Centre of Mass of a System of Particles

Observation.

Toss a spanner across a room. Each point of the spanner whirls in a complicated way, yet one special point glides along a smooth parabola, as if all the mass were concentrated there. That point is the **centre of mass**.

DEFINITION **Centre of mass**

The centre of mass of a system of particles of masses $m_1, m_2, \ldots$ at positions $\vec{r}_1, \vec{r}_2, \ldots$ is the mass-weighted average position

$$\vec{r}_{CM} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i} = \frac{\sum_i m_i \vec{r}_i}{M}, \qquad M = \sum_i m_i .$$

In coordinates: $x_{CM} = \dfrac{\sum m_i x_i}{M}$, and similarly for y_{CM} and z_{CM}.

WORKED EXAMPLE 15 **Centre of mass of two particles**

A 2 kg mass sits at $x = 0$ and a 6 kg mass at $x = 8$ m on a light rod. Locate the centre of mass.

SOLUTION

$$x_{CM} = \frac{2(0) + 6(8)}{2 + 6} = \frac{48}{8} = 6 \text{ m}.$$

The centre of mass is 6 m from the origin — closer to the heavier mass, as expected. Note the useful check: it divides the rod in the inverse ratio of the masses, $m_1 : m_2 = 1 : 3$, so it lies $\frac{3}{4}$ of the way from the lighter mass.

WORKED EXAMPLE 16 **Centre of mass of a simple array**

Three particles are placed as follows: 1 kg at $(0, 0)$, 2 kg at $(4, 0)$ and 3 kg at $(0, 6)$, coordinates in metres. Find the centre of mass.

SOLUTION

Total mass $M = 6$ kg. Then

$$x_{CM} = \frac{1(0) + 2(4) + 3(0)}{6} = \frac{8}{6} = \frac{4}{3} \text{ m}, \qquad y_{CM} = \frac{1(0) + 2(0) + 3(6)}{6} = \frac{18}{6} = 3 \text{ m}.$$

The centre of mass is at $\left(\frac{4}{3}, 3\right)$ m.

5.4 Motion of the Centre of Mass

Differentiate the definition of $\vec{r}_{CM}$ with respect to time:

$$M\vec{v}_{CM} = \sum_i m_i\vec{v}_i = \vec{P}_{\text{total}}.$$

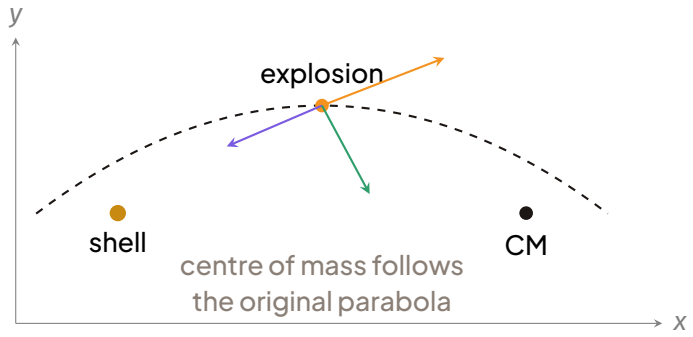
PROPERTY **Total momentum and the centre of mass**

The total momentum of a system equals the total mass times the velocity of the centre of mass:

$$\vec{P}_{\text{total}} = M\vec{v}_{CM}.$$

Consequently, for an **isolated system** (no external force), $\vec{P}_{\text{total}}$ is constant, so the centre of mass moves with constant velocity — in a straight line at constant speed, or remains at rest if it started at rest. This holds no matter how violently the parts of the system interact internally.

This is the deep link between explosions and the centre of mass: fragments fly apart, but the centre of mass continues exactly as if nothing had happened.



A projectile explodes in mid-flight; the fragments scatter, but the centre of mass continues along the original parabolic path (dashed).

WORKED EXAMPLE 17 **An exploding projectile**

A 12 kg projectile is moving horizontally at 20 m s^{-1} when it explodes into two fragments of 8 kg and 4 kg. Just after the explosion the 8 kg fragment is moving at 26 m s^{-1} in the original direction. Find (a) the velocity of the 4 kg fragment, (b) the velocity of the centre of mass after the explosion.

SOLUTION

(a) Momentum conservation along the line of motion:

$$12(20) = 8(26) + 4v \quad \Longrightarrow \quad 240 = 208 + 4v \quad \Longrightarrow \quad v = 8\,\text{m s}^{-1}.$$

(b) The centre of mass is unaffected by the internal explosion:

$$v_{CM} = \frac{8(26) + 4(8)}{12} = \frac{240}{12} = 20\,\text{m s}^{-1},$$

exactly the original velocity, as the property guarantees.

WARNING **Common mistakes with explosions and centre of mass**

- Applying elastic-collision rules (kinetic energy conserved) to an explosion — kinetic energy increases.
- Forgetting that momentum is a vector: fragments moving in different directions must be handled component by component.
- Thinking the centre of mass "jumps" when a body explodes: internal forces cannot move the centre of mass; only an external force can.
- In $F = v_{\text{rel}} \dfrac{dm}{dt}$, using the exhaust speed relative to the ground instead of relative to the rocket.

EXERCISE 4 **Explosion, rocket and centre of mass**

1. A stationary 5 kg body explodes into fragments of 2 kg and 3 kg. The 3 kg fragment moves at $10\,\text{m s}^{-1}$. Find the speed of the other fragment and the kinetic energy released.
2. A rocket ejects 120 kg of gas per second at a relative speed of $1800\,\text{m s}^{-1}$. Calculate the thrust, and the initial acceleration if the rocket's initial mass is 30000 kg (ignore gravity).
3. Particles of 4 kg at $(1, 2)$ and 6 kg at $(7, 8)$ (metres) are joined by a light rod. Find the centre of mass.
4. A 10 kg shell moving at $30\,\text{m s}^{-1}$ horizontally explodes into two equal fragments. One fragment stops dead. Find the velocity of the other and the velocity of the centre of mass.

SOLUTION **Exercise 1**

1. Momentum: $3(10) = 2v \Rightarrow v = 15\ \text{m s}^{-1}$ opposite to the 3 kg fragment. Energy released $= \frac{1}{2}(3)(10^2) + \frac{1}{2}(2)(15^2) = 150 + 225 = 375\ \text{J}$.
2. $F = v_{\text{rel}}\dfrac{dm}{dt} = 1800 \times 120 = 2.16e5\ \text{N}$; $a = \dfrac{F}{m} = \dfrac{2.16 \times 10^5}{3 \times 10^4} = 7.2\ \text{m s}^{-2}$.
3. $x_{CM} = \dfrac{4(1) + 6(7)}{10} = 4.6$, $y_{CM} = \dfrac{4(2) + 6(8)}{10} = 5.6$; centre of mass at $(4.6,\ 5.6)$ m.
4. Momentum: $10(30) = 5(0) + 5v \Rightarrow v = 60\ \text{m s}^{-1}$ in the original direction. Centre of mass: $v_{CM} = \dfrac{5(0) + 5(60)}{10} = 30\ \text{m s}^{-1}$, unchanged by the explosion.

KEY POINTS **Key points of this section**

- In an explosion, total momentum is conserved (zero if the body was at rest); kinetic energy increases by the chemical or stored energy released.
- Rocket thrust: $F = v_{\text{rel}}\dfrac{dm}{dt}$; rockets are variable-mass systems, and multi-staging sheds dead weight to boost acceleration.
- Centre of mass: $\vec{r}_{CM} = \dfrac{\sum m_i \vec{r}_i}{M}$; total momentum $\vec{P} = M\vec{v}_{CM}$.
- For an isolated system the centre of mass moves with constant velocity — fragments of an explosion scatter, but the centre of mass continues on the original path.

6 REAL-LIFE APPLICATIONS AND SYNTHESIS OF MOMENTUM AS A VECTOR

In the previous sections we saw how one single principle — the conservation of linear momentum — governs explosions, rocket propulsion and the motion of the centre of mass. We now close the chapter by returning to the everyday world: road accidents in Douala, a goalkeeper catching a shot, a fisherman pushing his pirogue on the Wouri. In every case the same two ideas do all the work:

$$\vec{J} = \vec{F}_{\text{av}}\,\Delta t = \Delta\vec{p} \qquad \text{and} \qquad \sum \vec{p}_{\text{before}} = \sum \vec{p}_{\text{after}} \quad \text{(isolated system)}.$$

This section trains you to recognise which idea applies, to use it quantitatively, and to present complete vector answers — exactly what the GCE Advanced Level examiner rewards. It directly addresses Lessons 198 (Force–Time Graphs/Application in Safety) and 202 (Real-life Applications of Momentum as a Vector) of the official syllabus.

6.1 Road Safety: Impact Forces from Momentum Change

When a vehicle of mass m moving at speed u is brought to rest, its momentum changes by $\Delta p = mu$ (in magnitude). If the stopping takes a time Δt, the average force on the vehicle (and, by Newton's third law, on whatever stops it) is

$$F_{\text{av}} = \frac{\Delta p}{\Delta t} = \frac{mu}{\Delta t}.$$

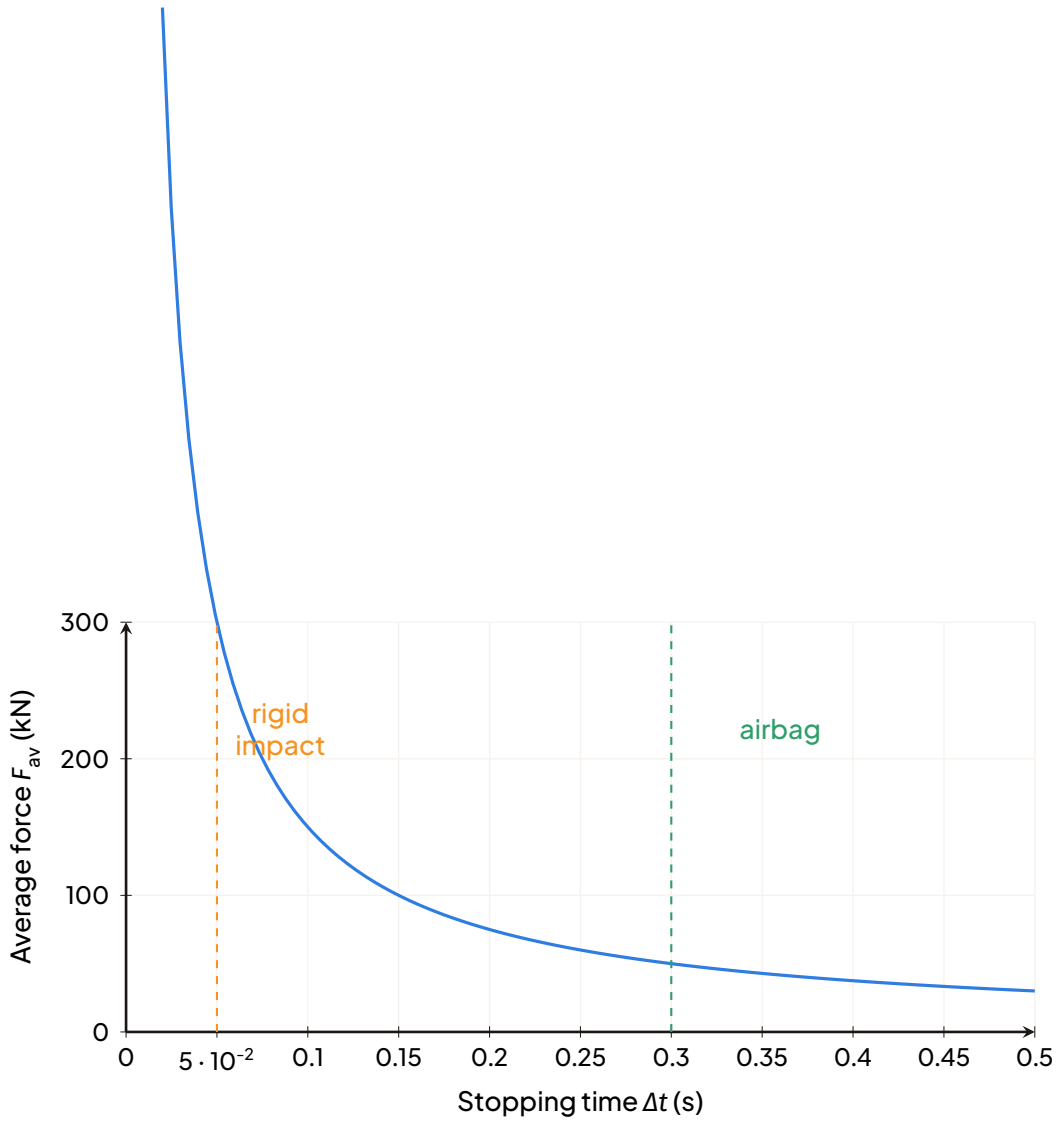
Two lessons follow immediately. First, F_{av} grows linearly with speed: doubling your speed doubles the momentum to be destroyed, and since a collision with a rigid obstacle stops the car in roughly the same (very short) time, the impact force doubles too. Second, F_{av} is inversely proportional to the stopping time: anything that lengthens Δt reduces the force.

PROPERTY **The core principle of all safety devices**

For a fixed change of momentum Δp, the average force is inversely proportional to the contact time:

$$F_{\text{av}} = \frac{\Delta p}{\Delta t} \propto \frac{1}{\Delta t}.$$

Every safety device — airbag, helmet, crumple zone, seatbelt, landing mat, boxing glove — works by increasing Δt, never by changing Δp.



Average impact force versus stopping time for a fixed momentum change $\Delta p = 15$ kN s: the hyperbola $F = \Delta p/\Delta t$. Stretching the contact time from 0.05 s to 0.3 s divides the force by six.

WORKED EXAMPLE 18 **Why speed limits save lives**

A car of mass 1000 kg carrying a driver of mass 70 kg travels at 90 km h^{-1} (= 25 m s^{-1}). In a head-on collision the car stops in 0.10 s. (a) Find the average force on the car. (b) The driver, restrained by a seatbelt and airbag, stops in 0.35 s; find the average force on the driver. (c) Comment on the effect of doubling the speed.

SOLUTION

Solution. (a) $\Delta p_{\text{car}} = 1000 \times 25 = 25\,000$ kg m s^{-1}, so

$$F_{\text{car}} = \frac{25\,000}{0.10} = 2.5 \times 10^5 \text{ N}.$$

(b) $\Delta p_{\text{driver}} = 70 \times 25 = 1750$ kg m s^{-1}, so

$$F_{\text{driver}} = \frac{1750}{0.35} = 5.0 \times 10^3 \text{ N},$$

about seven times the driver's weight — survivable when spread over the chest by the belt and airbag. Without the airbag the head might stop in 0.01 s against the steering wheel, giving forces ten times larger, concentrated on the skull. (c) At 180 km h^{-1} every Δp doubles, so every force above doubles. Speed limits matter because force — not just distance — grows with speed.

6.2 Safety Engineering and Sports: Impulse–Time Devices

All the devices studied qualitatively earlier can now be treated quantitatively with $F_{\text{av}} = \Delta p / \Delta t$:

- **Airbags and crumple zones** stretch the stopping time of the passenger and of the car respectively.
- **Helmets** contain a crushable foam liner: the head's momentum is destroyed over several milliseconds instead of a fraction of one.
- **Seatbelts** stretch slightly and hold the passenger to the car so that the passenger decelerates over the car's (longer) stopping time rather than against the dashboard.
- **Sports:** a cricketer or goalkeeper relaxes the hands and draws them back while catching, lengthening Δt; a footballer follows through the kick, lengthening the contact time so that a larger impulse $F\,\Delta t$ is transferred to the ball; high-jumpers land on thick mats; boxers wear padded gloves — all impulse–time devices.

WARNING **Following through versus catching**

Do not confuse the two directions of the same formula. Following through increases Δt to increase the impulse delivered ($\Delta p = F\Delta t$ grows). Catching with relaxed hands increases Δt to decrease the force received for a fixed Δp. State clearly which quantity is fixed in each situation.

6.3 Recoil and Propulsion in Daily Life

Recoil is conservation of momentum in a system that starts at rest — an "explosion" in the language of Section 5.

EXEMPLE **Everyday recoil and propulsion**

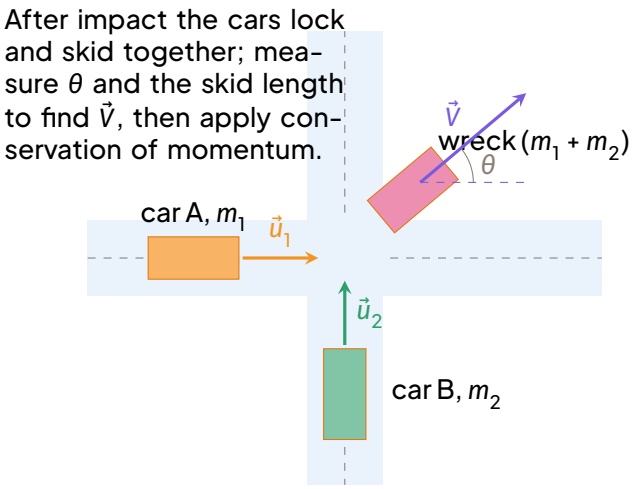
- Firing a gun: bullet forward, gun backward; $m_b v_b = m_g v_g$ in magnitude, so the heavy gun recoils slowly.
- Jet engine and rocket: hot gas ejected backward at high speed pushes the craft forward; thrust $F = v_e \left|\frac{dm}{dt}\right|$.
- Fire hose / water cannon: the hose must be held firmly because ejecting water of mass flow $\frac{dm}{dt}$ at speed v requires a backward reaction $v\frac{dm}{dt}$ on the hose.
- The pirogue trick: a fisherman stranded on a still pirogue throws a heavy load backward; he and the pirogue drift forward, since the total momentum of the (nearly isolated) system must remain zero.

6.4 Forensic Reconstruction of Collisions

Traffic investigators use conservation of momentum backwards: from the motion after a crash (skid distances, final positions) they estimate speeds before it. The impact itself is so brief that external impulses (friction during the few milliseconds of contact) are negligible, so momentum is conserved across the collision even when kinetic energy is not.

WARNING **Real collisions are never perfectly isolated**

Friction, braking forces and road gradients act during every collision. Momentum conservation is still an excellent approximation because the impact time is extremely short: the external impulse $\vec{F}_{\text{ext}}\Delta t$ is negligible compared with the momenta involved. Always state this justification in an exam solution — it is worth marks.



Junction collision reconstruction: car A approaches eastward, car B northward; the locked wreck moves off at angle θ. Momentum is conserved component by component.

WORKED EXAMPLE 19 **Estimating speeds before a junction collision**

Car A of mass 1200 kg travelling east and car B of mass 1000 kg travelling north collide at a junction and lock together. The wreck skids off at $\theta = 40^\circ$ north of east, and from the skid marks its speed just after impact is estimated at $V = 12\ \text{m s}^{-1}$. Estimate the speed of each car before impact.

SOLUTION

Solution. Take axes x east, y north. Momentum is conserved in each component (impact of very short duration).

$$x: \quad 1200\,u_1 = (1200 + 1000)(12\cos 40^\circ) \;\Rightarrow\; u_1 = \frac{2200 \times 12 \times 0.766}{1200} \approx 16.9\ \text{m s}^{-1}.$$

$$y: \quad 1000\,u_2 = 2200 \times 12 \sin 40^\circ \;\Rightarrow\; u_2 = \frac{2200 \times 12 \times 0.643}{1000} \approx 17.0\ \text{m s}^{-1}.$$

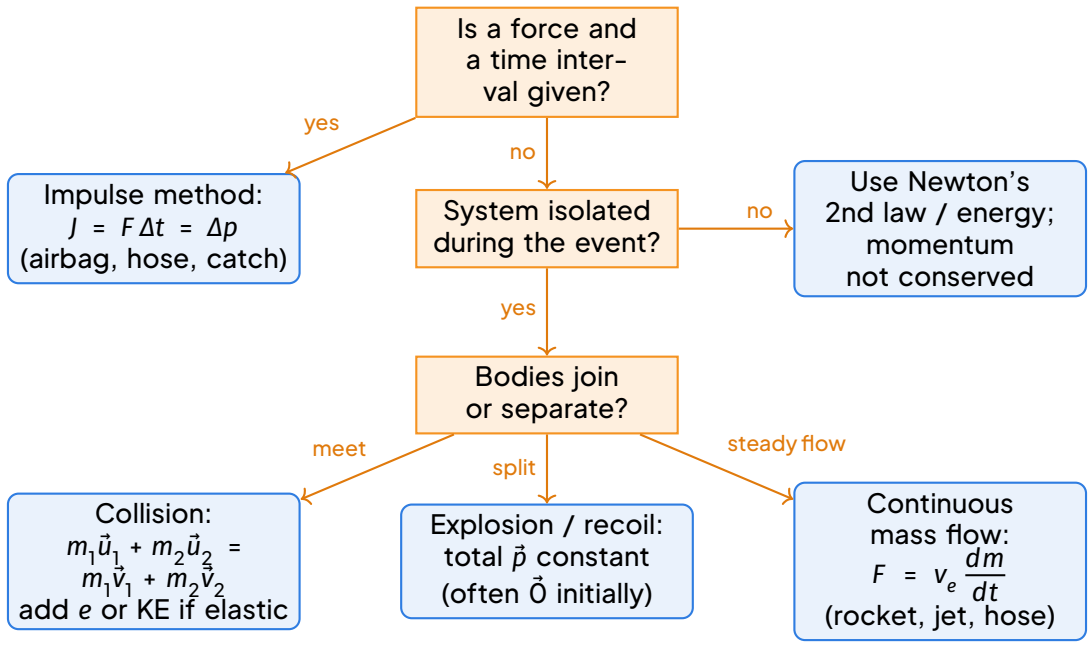
Both cars were travelling at about $61\ \text{km h}^{-1}$ — useful evidence if the limit was $50\ \text{km h}^{-1}$.

6.5 Choosing the Right Model: A Modelling Checklist

The hardest part of a momentum problem is rarely the algebra — it is recognising the type of situation.

METHOD **Modelling checklist for any momentum problem**

1. Define the system. Which bodies are included? Is it (approximately) isolated during the event?
2. Classify the interaction. Collision (bodies meet), explosion/recoil (one body splits or pushes off), or continuous impulse (force acting over a measurable time, e.g. hose, rocket, airbag)?
3. Choose the tool. Isolated collision/explosion ⇒ conservation of momentum (component by component in 2D). Force and time given ⇒ impulse $\vec{J} = \vec{F}\Delta t = \Delta\vec{p}$. Perfectly elastic ⇒ add conservation of kinetic energy (or $e = 1$).
4. List knowns and unknowns, fix a positive direction (or axes), and assign signs before substituting numbers.
5. Solve, then check: units, signs, and physical plausibility (e.g. a light object can rebound faster than it arrived, but kinetic energy must not increase in a collision).



Decision tree for selecting the correct momentum method.

6.6 Exam Strategy and Synthesis

WARNING **Presenting vector answers**

A momentum or velocity answer in two dimensions is never complete without both magnitude and direction, and units. Write, for example, "$|\vec{p}| = 26\,\text{kg m s}^{-1}$ at 23° north of east". Remember: $1\,\text{N s} = 1\,\text{kg m s}^{-1}$. In one dimension, a negative sign is the direction — state what it means ("the ball rebounds").

EXERCISE 5 **Synthesis: the runaway truck**

A truck of mass 3000 kg moving at $20\,\text{m s}^{-1}$ strikes a stationary car of mass 1000 kg; the two lock together. (a) Find their common speed just after impact. (b) They then skid to rest in 4.0 s under a constant friction force; find this force using impulse. (c) Show that kinetic energy is not conserved and find the fraction lost.

SOLUTION **Exercise 1**

(a) Conservation of momentum: $3000 \times 20 = 4000\,V \Rightarrow V = 15\,\text{m s}^{-1}$. (b) Impulse: $F\Delta t = \Delta p = 4000 \times 15 = 60\,000\,\text{kg m s}^{-1}$, so $F = \dfrac{60\,000}{4.0} = 1.5 \times 10^4\,\text{N}$. (c) KE before $= \frac{1}{2}(3000)(20^2) = 6.0 \times 10^5\,\text{J}$; KE after $= \frac{1}{2}(4000)(15^2) = 4.5 \times 10^5\,\text{J}$. Fraction lost $= \dfrac{1.5 \times 10^5}{6.0 \times 10^5} = 0.25$, i.e. 25%, converted to heat, sound and deformation.

KEY POINTS **One principle unifies the whole chapter**

- Impulse: $\vec{J} = \vec{F}_{\text{av}}\Delta t = \Delta\vec{p}$ — links force, time and momentum; the basis of all safety devices ($F_{\text{av}} \propto 1/\Delta t$).
- Conservation: for an isolated system, $\sum \vec{p}$ is constant — the same law solves collisions (elastic and inelastic, 1D and 2D), explosions, recoil and forensic reconstructions.
- Continuous flow: thrust $F = v_e \dfrac{dm}{dt}$ for rockets, jets and hoses.
- Centre of mass: the centre of mass of an isolated system moves with constant velocity — momentum conservation seen "from above".
- Always: define the system, justify isolation by the short impact time, fix axes and signs, and conclude with magnitude and direction, with units (kg m s^{-1} or N s).

DID YOU KNOW? **Momentum around us**

Crash-test programmes used by vehicle manufacturers worldwide are essentially large-scale momentum experiments: dummies instrumented with accelerometers measure $F\,\Delta t$ during impacts, and every improvement — collapsible steering columns, side-impact bars, helmet standards — is an application of $F_{\text{av}} = \Delta p / \Delta t$. The same equation that a goalkeeper uses instinctively when softening a catch guides the engineers designing the vehicles on the Yaoundé–Douala highway. One principle, countless lives saved.

7 INTEGRATION ACTIVITY

8 INVESTIGATIVE ACTIVITY: THE MILE 4 JUNCTION CRASH

8.1 Aims of the Activity

By the end of this group investigation, you should be able to:

- apply the **principle of conservation of linear momentum** in **two dimensions**, working component by component;
- compute the **kinetic energy** before and after a collision and use the result to **classify** the collision as elastic, inelastic or perfectly inelastic;
- use the **impulse–momentum relation** $F_{\text{av}}\,\Delta t = \Delta p$ to estimate average forces and to explain, quantitatively, how **safety devices** (seat belts, airbags, crumple zones) reduce injuries;
- communicate a complete scientific argument: set up axes, state assumptions, present calculations with units, and defend conclusions in front of the class;
- connect classroom mechanics to a genuine **road-safety concern** in Cameroon.

WARNING **Ground Rules**

Work in groups of four. Every numerical answer must carry its unit and be justified by a formula written in symbols first. Take $g = 9.81\ \text{m s}^{-2}$ where needed. Neglect friction during the impact itself (the collision time is very short, so external impulses during the impact are negligible).

8.2 Situation

The Mile 4 Junction Crash (Limbe).

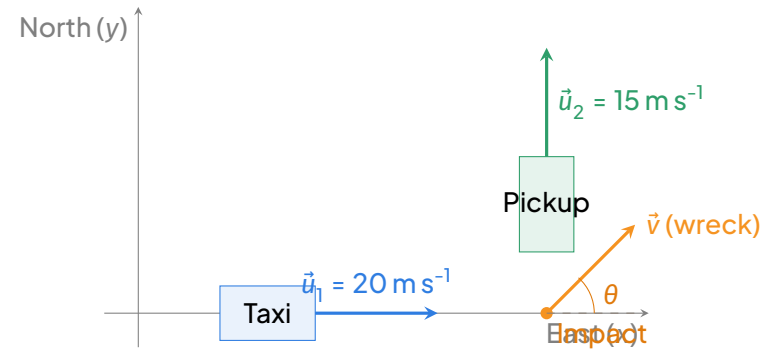
At 7:40 on a Monday morning, at a cross junction near Mile 4, a taxi (a saloon car) of mass $m_1 = 1200\ \text{kg}$ was travelling **due East** at $u_1 = 20\ \text{m s}^{-1}$ (about $72\ \text{km h}^{-1}$). At the same moment, a pickup truck of mass $m_2 = 1800\ \text{kg}$ was travelling **due North** at $u_2 = 15\ \text{m s}^{-1}$ (about $54\ \text{km h}^{-1}$). Neither driver stopped at the junction. The vehicles collided at right angles, **locked together** at impact, and the combined wreck skidded off along the tarmac.

A 70 kg passenger was sitting in the front seat of the taxi, moving with the taxi at $20\ \text{m s}^{-1}$ before the crash. Investigators consider two scenarios for this passenger during the collision:

- **Scenario A (no airbag, no effective restraint):** the passenger is brought to rest against the dashboard in about $\Delta t_A = 0.02\ \text{s}$;

- **Scenario B (seat belt and airbag):** the passenger is brought to rest over about $\Delta t_B = 0.15$ s.

The diagram below summarises the situation. Take the x-axis pointing East and the y-axis pointing North.



The taxi moves East, the pickup moves North; after locking together, the wreck moves with velocity $\vec{v}$ at angle θ North of East.

You are the team of **scientific investigators** attached to the case. Your report will be forwarded to the Divisional Delegation of Transport and used in a road-safety campaign.

8.3 Tasks

Part A — Velocity of the Wreck (Conservation of Momentum in 2D)

1. Write down the momentum vector of each vehicle **just before** the collision, giving the x-component (East) and y-component (North) separately, in $\mathrm{kg\,m\,s^{-1}}$.
2. State the principle of conservation of linear momentum and explain briefly why it applies here component by component, even though the collision is violent.
3. Hence find the components v_x and v_y of the velocity of the 3000 kg wreck just after impact.
4. Deduce the **magnitude** v of this velocity and the **angle** θ it makes with the East direction. Give θ to one decimal place.

Part B — Energy Analysis (Nature of the Collision)

5. Calculate the total kinetic energy of the two vehicles **before** the collision.
6. Calculate the kinetic energy of the wreck **after** the collision.
7. Determine the kinetic energy **lost** during the impact. Into what forms has this energy been transformed? State, with justification, whether the collision is elastic, inelastic or perfectly inelastic.

Part C — Impulse and the Passenger (Force–Time Reasoning)

8. Compute the change of momentum Δp of the 70 kg passenger, initially moving at $20\ \mathrm{m\,s^{-1}}$, who is brought to rest during the crash.

9. Using $F_{\text{av}} = \dfrac{\Delta p}{\Delta t}$, estimate the average force exerted on the passenger in **Scenario A** (Δt_A = 0.02 s) and in **Scenario B** (Δt_B = 0.15 s).
10. Express each force as a multiple of the passenger's weight ($W = mg$). Comment on the comparison.
11. Sketch, on the same axes, two force–time graphs (one per scenario) showing the same area under each curve. What does this area represent?

Part D — Communication

12. In at most ten lines, write a **safety recommendation** for the road-safety campaign, addressed to drivers and passengers in the South-West Region. Your text must use at least three of these words correctly: momentum, impulse, stopping time, conservation, kinetic energy.

Each group then presents Parts A–C on the board; the class challenges the assumptions and checks the units.

8.4 Model Answers and Examiner's Commentary

Part A

1. Momenta before impact (East = x, North = y):

$$\text{Taxi:}\quad p_{1x} = m_1 u_1 = 1200 \times 20 = 24\,000 \text{ kg m s}^{-1}, \qquad p_{1y} = 0,$$
$$\text{Pickup:}\quad p_{2x} = 0, \qquad p_{2y} = m_2 u_2 = 1800 \times 15 = 27\,000 \text{ kg m s}^{-1}.$$

2. Principle: If the resultant external force on a system is zero (or negligible during the event), the total linear momentum of the system remains constant. During the very short impact, the mutual collision forces are enormous compared with friction from the road, so external impulse is negligible. Since momentum is a **vector**, the principle holds separately along x and along y.

3. Let $\vec{v} = (v_x, v_y)$ be the velocity of the combined wreck of mass $M = 1200 + 1800 = 3000$ kg:

$$x\text{-axis:}\quad 24\,000 = 3000\,v_x \;\Rightarrow\; v_x = 8 \text{ m s}^{-1},$$
$$y\text{-axis:}\quad 27\,000 = 3000\,v_y \;\Rightarrow\; v_y = 9 \text{ m s}^{-1}.$$

4. Magnitude and direction:

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{8^2 + 9^2} = \sqrt{145} \approx 12.0 \text{ m s}^{-1}, \qquad \tan\theta = \frac{v_y}{v_x} = \frac{9}{8} \;\Rightarrow\; \theta \approx 48.4^\circ \text{ North of East.}$$

Part B

5. Kinetic energy before:

$$E_{k,\text{before}} = \tfrac{1}{2}(1200)(20)^2 + \tfrac{1}{2}(1800)(15)^2 = 240\,000 + 202\,500 = 442\,500 \text{ J}.$$

6. Kinetic energy after:

$$E_{k,\text{after}} = \tfrac{1}{2}(3000)\left(\sqrt{145}\right)^2 = \tfrac{1}{2}(3000)(145) = 217\,500 \text{ J}.$$

7. Energy lost:

$$\Delta E_k = 442\,500 - 217\,500 = 225\,000\ \text{J} = 225\ \text{kJ}.$$

This energy is transformed into **deformation** of the bodywork (permanent bending of metal), **sound**, and **heat**. Since kinetic energy is not conserved and the two vehicles move together after impact, the collision is **perfectly inelastic**. Note that about 51% of the initial kinetic energy is dissipated — a perfectly inelastic collision is the most dissipative type consistent with momentum conservation.

Part C

8. Change of momentum of the passenger (brought from $20\ \text{m s}^{-1}$ to rest):

$$\Delta p = m\,\Delta u = 70 \times 20 = 1400\ \text{kg m s}^{-1}.$$

9. Average forces:

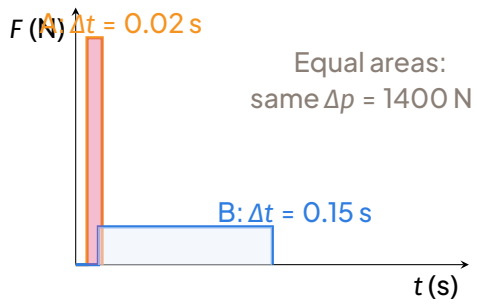
$$F_A = \frac{1400}{0.02} = 70\,000\ \text{N}, \qquad F_B = \frac{1400}{0.15} \approx 9\,333\ \text{N}.$$

10. The passenger's weight is $W = mg = 70 \times 9.81 \approx 687$ N. Hence

$$\frac{F_A}{W} \approx 102, \qquad \frac{F_B}{W} \approx 13.6.$$

Without restraint, the passenger experiences a force about **one hundred times his own weight**, which is almost certainly fatal. With seat belt and airbag, the same change of momentum is spread over a stopping time 7.5 times longer, so the average force is divided by 7.5: the impulse $F\,\Delta t = \Delta p$ is fixed by the momentum change, only its distribution in time changes.

11. Force–time sketch:



Both force–time pulses enclose the same area $\int F\,dt = \Delta p = 1400\ \text{kg m s}^{-1}$; the airbag lowers the peak force by lengthening the stopping time.

The area under each curve is the **impulse**, equal to the change of momentum of the passenger: identical in both scenarios.

Part D (Sample Recommendation)

"Every moving vehicle carries momentum, and in a crash that momentum must be removed by an impulse. Physics cannot reduce the impulse — your change of momentum is fixed by your speed — but it can stretch the stopping time. Seat belts and airbags increase the stopping time of your body, dividing the average force by a large factor and saving lives. Drivers: reduce speed, because both momentum and kinetic energy grow with speed. Passengers: fasten your seat belt on every journey, even short ones. Let conservation of momentum work for you, not against you."

KEY POINTS **What This Activity Consolidates**

- Momentum is a vector: in two dimensions, conserve it component by component.
- Locking together ⇒ perfectly inelastic: momentum conserved, kinetic energy not.
- $F_{av}\,\Delta t = \Delta p$: for a fixed Δp, a longer stopping time means a smaller average force — the principle behind every safety device.

9 METHODS AND WORKED EXAMPLES

9.1 Method 1: Computing Momentum and Impulse as Vectors

DEFINITION **Momentum and impulse**

The **momentum** of a body of mass m moving with velocity $\vec{v}$ is the vector

$$\vec{p} = m\vec{v}, \qquad \text{unit: kg m s}^{-1} \text{ (equivalently N s).}$$

The **impulse** of a force $\vec{F}$ acting from time t_1 to t_2 is

$$\vec{J} = \vec{F}\,\Delta t \text{ (constant force)}, \qquad \vec{J} = \int_{t_1}^{t_2} \vec{F}\,dt \text{ (variable force).}$$

PROPERTY **Impulse–momentum theorem**

The impulse of the resultant force acting on a body equals its change of momentum:

$$\vec{J} = \Delta\vec{p} = m\vec{v} - m\vec{u}.$$

This follows directly from Newton's second law $\vec{F} = \dfrac{d\vec{p}}{dt}$ integrated over time. (The derivation from $\vec{F} = m\vec{a}$ for constant mass is examinable.)

METHOD **Momentum and impulse of a moving body**

1. Identify the mass m (in kg) and write the velocity as a **vector** $\vec{v}$ (choose unit vectors $\vec{i}, \vec{j}$ or a sign convention in 1D).
2. Compute the momentum $\vec{p} = m\vec{v}$. Its unit is kg m s^{-1} (equivalently N s).
3. For a change of velocity, compute $\Delta\vec{p} = m\vec{v} - m\vec{u}$ vectorially: subtract components, never magnitudes blindly.
4. The impulse of a force is $\vec{J} = \vec{F}\,\Delta t$ (constant force) or $\vec{J} = \int_{t_1}^{t_2} \vec{F}\,dt$; the impulse–momentum theorem gives $\vec{J} = \Delta\vec{p}$.
5. If the force varies, the impulse is the **area under the force–time graph**.

Reflex: momentum and impulse are vectors — always state a positive direction and keep

signs consistent.

WORKED EXAMPLE 20 **Impulse on a football**

A football of mass 0.45 kg is moving horizontally with velocity $(12\vec{i} + 5\vec{j})\ \text{m s}^{-1}$ when a player kicks it. Immediately after the kick its velocity is $(-8\vec{i} + 14\vec{j})\ \text{m s}^{-1}$.

(a) Find the momentum of the ball before and after the kick.

(b) Find the impulse imparted to the ball by the kick, and state its magnitude.

(c) The contact between boot and ball lasts 0.08 s. Find the average force exerted on the ball.

SOLUTION

Solution. Take $\vec{i}, \vec{j}$ as fixed horizontal unit vectors.
(a) Momentum is $\vec{p} = m\vec{v}$ with $m = 0.45$ kg.

$$\vec{p}_{\text{before}} = 0.45(12\vec{i} + 5\vec{j}) = (5.4\vec{i} + 2.25\vec{j})\ \text{kg m s}^{-1}.$$

$$\vec{p}_{\text{after}} = 0.45(-8\vec{i} + 14\vec{j}) = (-3.6\vec{i} + 6.3\vec{j})\ \text{kg m s}^{-1}.$$

(b) By the impulse–momentum theorem,

$$\begin{aligned}\vec{J} = \Delta\vec{p} &= \vec{p}_{\text{after}} - \vec{p}_{\text{before}} \\ &= (-3.6 - 5.4)\vec{i} + (6.3 - 2.25)\vec{j} \\ &= (-9\vec{i} + 4.05\vec{j})\ \text{N s}.\end{aligned}$$

Its magnitude is

$$|\vec{J}| = \sqrt{9^2 + 4.05^2} = \sqrt{81 + 16.4025} = \sqrt{97.4025} \approx 9.87\ \text{N s}.$$

(c) The average force satisfies $\vec{J} = \vec{F}_{\text{av}}\Delta t$, so

$$\vec{F}_{\text{av}} = \frac{\vec{J}}{\Delta t} = \frac{-9\vec{i} + 4.05\vec{j}}{0.08} = (-112.5\vec{i} + 50.6\vec{j})\ \text{N}.$$

Magnitude: $|\vec{F}_{\text{av}}| = \dfrac{9.87}{0.08} \approx 123\ \text{N}$.
Check: the impulse points roughly opposite to the initial $\vec{i}$-motion (the ball is reversed in the $\vec{i}$-direction) and in the positive $\vec{j}$-direction — consistent with the observed change of velocity.

9.2 Method 2: Applying Conservation of Linear Momentum

PROPERTY **Principle of conservation of linear momentum**

If the resultant external force on a system is zero (or its impulse during the event is negligible), then the total momentum of the system is constant:

$$\sum m_i \vec{u}_i = \sum m_i \vec{v}_i.$$

Internal forces between the bodies (collision or explosion forces) occur in equal and opposite pairs by Newton's third law, so they cancel in the total. (The justification from Newton's second and third laws is examinable.)

METHOD **Conservation of momentum in 1D**

1. Define the **system** (the colliding or separating bodies) and check that external impulses during the event are negligible (short collision, smooth surface).
2. Choose a positive direction and write the velocity of each body with its sign.
3. Write total momentum before = total momentum after:

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2.$$

4. Solve for the unknown. A negative answer means motion opposite to the chosen positive direction.
5. For bodies that coalesce (stick together), use a single combined mass $(m_1 + m_2)$ after impact.

Reflex: draw a "before" and an "after" diagram with arrows — most sign errors disappear.

WORKED EXAMPLE 21 **Trucks coupling on a straight rail**

A railway truck A of mass 6000 kg moving at 4 m s^{-1} on a straight horizontal track collides with a stationary truck B of mass 4000 kg. On impact the two trucks couple together and move as one.

(a) Find their common speed just after the collision.

(b) Find the kinetic energy lost in the collision and comment on the nature of the collision.

SOLUTION

Solution. Take the direction of motion of A as positive. The track is horizontal and the collision is brief, so momentum is conserved.

(a) Before: $u_A = 4$, $u_B = 0$. After: combined mass $6000 + 4000 = 10000$ kg moving at speed v.

$$m_A u_A + m_B u_B = (m_A + m_B)v$$
$$6000(4) + 4000(0) = 10000\,v$$
$$24000 = 10000\,v \quad \Longrightarrow \quad v = 2.4\,\text{m s}^{-1}.$$

The trucks move together at $2.4\,\text{m s}^{-1}$ in the original direction of A.
(b) Kinetic energy before:

$$KE_{\text{before}} = \tfrac{1}{2}(6000)(4^2) + 0 = 48000\,\text{J}.$$

Kinetic energy after:

$$KE_{\text{after}} = \tfrac{1}{2}(10000)(2.4^2) = \tfrac{1}{2}(10000)(5.76) = 28800\,\text{J}.$$

Loss: $48000 - 28800 = 19200$ J.
Since kinetic energy is not conserved, the collision is **inelastic** (perfectly inelastic, since the bodies stick). The lost energy becomes heat, sound and permanent deformation of the coupling mechanism.

9.3 Method 3: Elastic Collisions in One Dimension

METHOD **Solving a 1D elastic collision**

1. Write two equations: conservation of momentum

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

and conservation of kinetic energy (or, quicker, Newton's experimental law with $e = 1$):

$$v_2 - v_1 = u_1 - u_2 \quad \text{(speed of separation = speed of approach)}.$$

2. Solve the simultaneous linear system for v_1 and v_2.
3. Useful memorised results for a target at rest ($u_2 = 0$):

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2}u_1, \qquad v_2 = \frac{2m_1}{m_1 + m_2}u_1.$$

4. Interpret signs: $v_1 < 0$ means the incoming body rebounds.

Reflex: for an elastic collision never use energy conservation alone — you need both equations.

WORKED EXAMPLE 22 **Elastic collision between two spheres**

A sphere P of mass 2 kg moving at 6 m s^{-1} collides directly with a sphere Q of mass 3 kg moving at 1 m s^{-1} in the same direction on a smooth horizontal table. The collision is perfectly elastic. Find the velocities of P and Q after the collision.

SOLUTION

Solution. Positive direction: direction of motion of P. Given $u_P = 6$, $u_Q = 1$.
Momentum:

$$2(6) + 3(1) = 2v_P + 3v_Q \quad \Longrightarrow \quad 2v_P + 3v_Q = 15. \tag{1}$$

Elastic condition (separation speed = approach speed):

$$v_Q - v_P = u_P - u_Q = 6 - 1 = 5. \tag{2}$$

From (2): $v_Q = v_P + 5$. Substitute in (1):

$$\begin{aligned} 2v_P + 3(v_P + 5) &= 15 \\ 5v_P + 15 &= 15 \\ v_P = 0, \qquad v_Q &= 5 \text{ m s}^{-1}. \end{aligned}$$

Answer: P is brought to rest; Q moves on at 5 m s^{-1} in the positive direction.
Verification. Momentum after: $2(0) + 3(5) = 15$ ✓. KE before: $\frac{1}{2}(2)(36) + \frac{1}{2}(3)(1) = 36 + 1.5 = 37.5$ J. KE after: $0 + \frac{1}{2}(3)(25) = 37.5$ J ✓. Both conserved, as required for an elastic collision.

9.4 Method 4: Newton's Law of Restitution (General 1D Impact)

METHOD **Impact with coefficient of restitution e**

1. Write conservation of momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.
2. Write **Newton's experimental law**:

   $$v_2 - v_1 = -e\,(u_2 - u_1) \quad \Longleftrightarrow \quad \text{speed of separation} = e \times \text{speed of approach},$$

   with $0 \le e \le 1$ ($e = 1$ elastic, $e = 0$ perfectly inelastic).
3. Solve the pair of linear equations.
4. For impact with a fixed wall, $u_2 = v_2 = 0$, giving simply $v_1 = -e\,u_1$ (rebound speed = $e\times$ approach speed).
5. Energy lost = $\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \frac{1}{2}m_1v_1^2 - \frac{1}{2}m_2v_2^2$.

WARNING **Sign trap in the restitution law**

The law relates speeds of separation and approach: always write it as $v_2 - v_1 = e(u_1 - u_2)$ with body 2 initially ahead of body 1. Reversing the order flips signs and ruins the solution.

WORKED EXAMPLE 23 **Ball rebounding from a wall**

A ball of mass 0.5 kg moves at 10 m s^{-1} towards a fixed vertical wall. The coefficient of restitution between ball and wall is $e = 0.6$.

(a) Find the velocity of the ball immediately after impact.

(b) Find the impulse exerted by the wall on the ball.

(c) Find the kinetic energy lost in the impact.

SOLUTION

Solution. Take the initial direction of the ball as positive: $u = 10$ m s^{-1}.
(a) For a fixed wall, rebound speed = $e\times$ approach speed, direction reversed:

$$v = -eu = -0.6(10) = -6 \text{ m s}^{-1}.$$

The ball rebounds at 6 m s^{-1} away from the wall.
(b) Impulse = change of momentum:

$$J = m(v - u) = 0.5(-6 - 10) = 0.5(-16) = -8 \text{ N s}.$$

The wall exerts an impulse of magnitude 8 N s on the ball, directed away from the wall.
(c)

$$\Delta KE = \tfrac{1}{2}mu^2 - \tfrac{1}{2}mv^2 = \tfrac{1}{2}(0.5)(100 - 36) = 0.25 \times 64 = 16 \text{ J}.$$

So 16 J is dissipated (sound, heat, deformation) in the impact.
Remark: note that the impulse involves the sum $u + |v|$ because the velocity changes sign — a very common source of error is to write $J = m(u - v)$ with unsigned speeds.

9.5 Method 5: Collisions in Two Dimensions

METHOD **2D collision problems**

1. Draw the situation: line of centres at impact, directions of motion before and after.

2. Apply conservation of momentum **as a vector equation**: write one equation per axis ($\vec{i}$ and $\vec{j}$ components separately).
3. For a smooth sphere striking a smooth fixed surface: the velocity component parallel to the surface is unchanged; the component perpendicular to the surface is multiplied by $-e$.
4. For oblique impact of two smooth spheres: momentum is conserved along the line of centres (with restitution e along that line), and the velocity of each sphere perpendicular to the line of centres is unchanged.
5. Use $\tan\theta = \dfrac{\text{perpendicular component}}{\text{parallel component}}$ for deflection angles.

Reflex: resolve before writing any equation; never apply restitution along a direction other than the line of centres.

WORKED EXAMPLE 24 **Oblique impact with a smooth wall**

A smooth sphere of mass 0.4 kg moves on a smooth horizontal floor and strikes a smooth vertical wall. Just before impact its velocity is $(4\vec{i} + 3\vec{j})\ \text{m s}^{-1}$, where $\vec{i}$ is perpendicular to the wall (pointing towards it) and $\vec{j}$ is parallel to the wall. The coefficient of restitution is $e = \frac{1}{2}$.

(a) Find the velocity of the sphere after the impact.

(b) Find the angle through which the direction of motion is deflected.

(c) Find the impulse exerted by the wall on the sphere.

SOLUTION

Solution.

(a) Component parallel to the wall ($\vec{j}$) is unchanged: $v_j = 3$. Component perpendicular to the wall is reversed and multiplied by e: $v_i = -e\,u_i = -\frac{1}{2}(4) = -2$.

$$\vec{v} = (-2\vec{i} + 3\vec{j})\ \text{m s}^{-1}.$$

(b) Before impact the velocity makes angle α with the wall where $\tan\alpha = \dfrac{u_i}{u_j} = \dfrac{4}{3}$, so $\alpha \approx 53.1^\circ$. After impact the angle with the wall is β with $\tan\beta = \dfrac{|v_i|}{v_j} = \dfrac{2}{3}$, so $\beta \approx 33.7^\circ$. The total deflection is

$$\alpha + \beta \approx 53.1^\circ + 33.7^\circ = 86.8^\circ.$$

(The incoming and outgoing paths lie on opposite sides of the normal, so the deflection is the sum.)

(c) Only the $\vec{i}$-component of momentum changes:

$$\vec{J} = m(\vec{v} - \vec{u}) = 0.4\left[(-2 - 4)\vec{i} + (3 - 3)\vec{j}\right] = -2.4\,\vec{i}\ \text{N s}.$$

The impulse has magnitude 2.4 N s and is directed along the normal, away from the wall — as expected, since a smooth wall can only push along its normal.

9.6 Method 6: Explosions, Recoil and Rocket Motion

METHOD **Explosions and recoil**

1. An explosion is a collision in reverse: total momentum is conserved (often zero before, hence zero after).
2. If the system starts at rest, write $m_1\vec{v}_1 + m_2\vec{v}_2 = \vec{0}$, so the fragments move in opposite directions with momenta of equal magnitude: $m_1 v_1 = m_2 v_2$.
3. Kinetic energy increases: the chemical (or elastic) energy released equals the gain in KE.
4. For a **rocket**: thrust $= u_{\text{rel}}\left|\dfrac{dm}{dt}\right|$, where u_{rel} is the exhaust speed relative to the rocket; apply $F = ma$ with the instantaneous mass.

Reflex: in recoil problems the lighter fragment always gets the larger speed.

WORKED EXAMPLE 25 **Recoil of a gun and rocket thrust**

Part A. A rifle of mass 4 kg fires a bullet of mass 20 g horizontally. The bullet leaves the muzzle at 400 m s^{-1}. Find the recoil speed of the rifle and the total kinetic energy generated by the explosion of the charge.
Part B. A rocket of instantaneous mass 1200 kg ejects gas at 1800 m s^{-1} (relative to the rocket) at a rate of 8 kg s^{-1}. Find the thrust and the initial acceleration of the rocket in deep space (no external force).

SOLUTION

Part A. System: rifle + bullet, initially at rest, so total momentum before = 0.
Conservation of momentum (taking the bullet's direction as positive):

$$0 = m_b v_b + m_r v_r \quad\Longrightarrow\quad 0 = (0.020)(400) + 4\,v_r.$$

$$v_r = -\frac{8}{4} = -2\ \text{m s}^{-1}.$$

The rifle recoils at 2 m s^{-1} opposite to the bullet.
Kinetic energy generated:

$$KE = \tfrac{1}{2}(0.020)(400)^2 + \tfrac{1}{2}(4)(2)^2 = 1600 + 8 = 1608\ \text{J}.$$

This energy comes from the chemical energy of the charge. Note the bullet carries $1600/1608 \approx 99.5\%$ of the kinetic energy, although the momenta of rifle and bullet have equal magnitudes.

Part B. Thrust:

$$F = u_{\text{rel}} \left| \frac{dm}{dt} \right| = 1800 \times 8 = 14400 \text{ N}.$$

Newton's second law with instantaneous mass $m = 1200$ kg:

$$a = \frac{F}{m} = \frac{14400}{1200} = 12 \text{ m s}^{-2}.$$

As fuel burns, m decreases while the thrust stays constant, so the acceleration increases with time — this is why rocket acceleration is not uniform.

9.7 Method 7: Force–Time Graphs and Safety Applications

METHOD **Reading and using F–t graphs**

1. The impulse equals the **area** between the F–t curve and the time axis: $J = \int F\,dt$ (triangles, rectangles or trapezia for straight-line graphs; integration for curved graphs).
2. Set this area equal to the change of momentum $m\Delta v$ to find an unknown speed, mass or time.
3. The average force is $\bar{F} = \dfrac{J}{\Delta t} = \dfrac{\text{area}}{\text{base}}$.
4. Safety principle: for a fixed change of momentum, increasing the impact time Δt decreases the peak and average force. This explains crumple zones, airbags, seatbelts, crash helmets, bending the knees on landing, and follow-through in sport.

WORKED EXAMPLE 26 **Airbag force–time graph**

In a road accident test, a dummy's head of mass 5 kg travelling at 15 m s^{-1} is brought to rest by an airbag. The force F (in N) exerted on the head varies with time t (in s) according to

$$F(t) = 450000\,t\,(0.1 - t), \qquad 0 \le t \le 0.1.$$

(a) Sketch the force–time graph and find the maximum force.

(b) Show that the impulse delivered by the airbag equals the change of momentum of the head.

(c) Without the airbag the head would be stopped by the dashboard in 0.01 s with approximately constant force. Compare the forces in the two cases and comment.

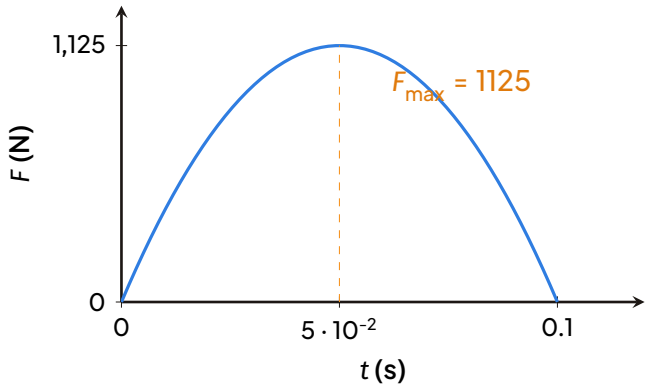
SOLUTION

Solution.

(a) Expanding, $F(t) = 45000t - 450000t^2$: a downward parabola with $F(0) = F(0.1) = 0$. The maximum occurs where

$$F'(t) = 45000 - 900000t = 0 \quad \Longrightarrow \quad t = 0.05\ \text{s}.$$

$$F_{\max} = 450000(0.05)(0.1 - 0.05) = 450000 \times 0.0025 = 1125\ \text{N}.$$



Force–time curve during airbag contact (parabolic model). The shaded region under the curve is the impulse.

(b) Impulse = area under the curve:

$$J = \int_0^{0.1} (45000t - 450000t^2)\,dt = \left[22500t^2 - 150000t^3\right]_0^{0.1}$$
$$= 22500(0.01) - 150000(0.001) = 225 - 150 = 75\ \text{N s}.$$

Change of momentum of the head (brought from $15\ \text{m s}^{-1}$ to rest):

$$\Delta p = m\,\Delta v = 5 \times 15 = 75\ \text{kg m s}^{-1} = 75\ \text{N s}.$$

Hence $J = \Delta p$: the impulse–momentum theorem is verified. (Quick check: the area of a parabola is $\frac{2}{3} \times$ base $\times$ height $= \frac{2}{3}(0.1)(1125) = 75$ N s ✓.)

(c) Without the airbag, a constant force acting for 0.01 s must deliver the same impulse 75 N s:

$$\bar{F} = \frac{J}{\Delta t} = \frac{75}{0.01} = 7500\ \text{N}.$$

With the airbag the peak force is only 1125 N. Multiplying the stopping time by 10 has divided the force by a factor of about 7: this is exactly the principle of the airbag — same change of momentum, longer time, smaller force — and it is why airbags, seatbelts and crumple zones save lives on Cameroonian roads as everywhere else.

9.8 Method 8: Centre of Mass and Momentum of a System

METHOD **Centre of mass and total momentum**

1. For particles m_i at positions $\vec{r}_i$: $\vec{r}_{CM} = \dfrac{\sum m_i \vec{r}_i}{\sum m_i}$; in 1D, $\bar{x} = \dfrac{\sum m_i x_i}{\sum m_i}$.
2. The total momentum of the system equals the momentum of a single particle carrying the total mass at the centre of mass:

$$\vec{P} = M\vec{v}_{CM}.$$

3. If no external force acts, $\vec{v}_{CM}$ is constant — even during collisions or explosions between the parts.
4. Use this to find the motion of one fragment when the other(s) and the initial velocity are known.

WORKED EXAMPLE 27 **Exploding shell and the centre of mass**

A shell of mass 6 kg is fired and, at the top of its flight, is moving horizontally at 50 m s^{-1} when it explodes into two fragments of masses 4 kg and 2 kg. The 4 kg fragment is brought to rest (momentarily) by the explosion. Air resistance is neglected.

(a) Find the velocity of the 2 kg fragment just after the explosion.

(b) Verify that the velocity of the centre of mass is unchanged by the explosion.

(c) Find the kinetic energy released by the explosion.

SOLUTION

Solution. Take the direction of the shell's motion as positive. Before explosion: $M = 6$ kg, $u = 50$ m s^{-1}.

(a) Momentum is conserved (the explosion forces are internal; gravity acts vertically and has negligible impulse during the instant of explosion):

$$\begin{aligned} Mu &= m_1 v_1 + m_2 v_2 \\ 6(50) &= 4(0) + 2v_2 \\ 300 &= 2v_2 \quad \Longrightarrow \quad v_2 = 150 \text{ m s}^{-1}. \end{aligned}$$

The 2 kg fragment shoots forward at 150 m s^{-1}.

(b) Velocity of centre of mass after the explosion:

$$v_{CM} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{4(0) + 2(150)}{6} = \frac{300}{6} = 50 \text{ m s}^{-1}.$$

This equals the shell's velocity before the explosion: the centre of mass continues exactly as if nothing had happened, and it would follow the original parabolic trajectory until a fragment hits the ground.
(c) Kinetic energy before:

$$KE_{\text{before}} = \tfrac{1}{2}(6)(50)^2 = 7500 \text{ J}.$$

Kinetic energy after:

$$KE_{\text{after}} = \tfrac{1}{2}(4)(0)^2 + \tfrac{1}{2}(2)(150)^2 = 22500 \text{ J}.$$

Energy released by the explosion:

$$\Delta KE = 22500 - 7500 = 15000 \text{ J},$$

supplied by the chemical energy of the explosive charge.
Remark: note the contrast with collisions — in an explosion kinetic energy increases, while momentum is still conserved. Momentum conservation is the universal tool; energy bookkeeping tells you the type of event.

KEY POINTS **Key points of the chapter**

- Momentum $\vec{p} = m\vec{v}$ and impulse $\vec{J} = \int \vec{F}\, dt$ are **vectors**; $\vec{J} = \Delta\vec{p}$ always.
- Total momentum is conserved whenever the external impulse is negligible — in collisions and in explosions.
- Kinetic energy is conserved only in **elastic** collisions ($e = 1$); it decreases in inelastic collisions ($e < 1$) and increases in explosions.
- Newton's experimental law: speed of separation = $e\times$ speed of approach, along the line of centres (or the normal to a wall).
- In 2D, resolve: momentum is conserved component by component; perpendicular components to a smooth wall are unchanged.
- Rocket thrust = $u_{\text{rel}}\left|\frac{dm}{dt}\right|$; the centre of mass of an isolated system moves with constant velocity.
- Safety devices work by lengthening the impact time: same $\Delta\vec{p}$, smaller force.

10 EXERCISES

10.1 I check my knowledge

EXERCISE 6 **Multiple choice: the nature of momentum**

Choose the correct answer and justify briefly.

1. The momentum of a body of mass m moving with velocity $\vec{v}$ is:

   (a) a scalar equal to mv;

   (b) a vector $\vec{p} = m\vec{v}$, parallel to $\vec{v}$;

   (c) a vector $m\vec{v}$, perpendicular to $\vec{v}$;

   (d) a scalar equal to $\frac{1}{2}mv^2$.

2. A 2 kg stone and a 1 kg stone have the same kinetic energy. Which has the greater momentum?

   (a) the 2 kg stone;

   (b) the 1 kg stone;

   (c) they are equal;

   (d) impossible to tell.

EXERCISE 7 **True or false**

State whether each statement is true or false, and justify your answer.

1. If the total external force on a system is zero, the total momentum of the system is constant.

2. In a perfectly inelastic collision, kinetic energy is conserved.

3. The impulse of a force equals the area under the force–time graph.

4. During an explosion of a stationary body, the fragments always move in the same direction.

EXERCISE 8 **Definitions**

Define each of the following, giving the SI unit of each quantity:

1. the linear momentum of a particle;
2. the impulse of a force;
3. an elastic collision;
4. the centre of mass of a system of particles.

EXERCISE 9 **Direct calculations**

1. A lorry of mass 8000 kg travels along a straight road at 15 m s^{-1}. Calculate its momentum.
2. A tennis ball of mass 0.058 kg moving at 30 m s^{-1} is stopped by a racket in 0.05 s. Find the average force exerted on the ball.
3. A constant force of 40 N acts on a body for 0.25 s. Find the impulse delivered to the body.

10.2 I practise

EXERCISE 10 **Kicking a football**

A 0.45 kg football moving at 18 m s^{-1} is kicked back along the same line at 24 m s^{-1}; the foot is in contact with the ball for 0.08 s.

1. Find the impulse given to the ball.
2. Find the average force exerted on the ball.

EXERCISE 11 **Man and canoe**

A 60 kg man jumps horizontally at 3 m s^{-1} from a stationary 90 kg canoe.

1. Find the recoil speed of the canoe.
2. State the physical principle you have used, and explain why it applies here.

EXERCISE 12 **Trolleys that stick together**

A 2 kg trolley moving at 5 m s^{-1} collides head-on with a 3 kg trolley moving at 1 m s^{-1} in the opposite direction; the trolleys stick together after impact.

1. Find their common velocity just after the collision.
2. Find the percentage of kinetic energy lost in the collision.

EXERCISE 13 **Force–time graph**

A force acting on a body varies with time as shown on a force–time graph consisting of a triangle of base 0.4 s and peak value 500 N.

1. Find the impulse of the force.
2. Find the change in velocity of a 4 kg body, initially at rest, on which this force acts.

10.3 I prepare for the GCE

EXERCISE 14 **Collision at a crossroads** **(12 marks)**

A 1500 kg car travelling east at 25 m s^{-1} collides at a crossroads with a 2000 kg truck travelling north at 15 m s^{-1}. The vehicles lock together on impact.

1. State the principle you will use and the conditions under which it is valid. (2)
2. Write down the components of the total momentum of the system before the collision along the east and north directions. (3)
3. Find the speed of the wreck immediately after the collision. (3)
4. Find the direction of motion of the wreck, giving the angle it makes with the east direction. (2)
5. Calculate the kinetic energy lost in the collision and comment on the nature of this collision. (2)

EXERCISE 15 **Ballistic pendulum and airbag safety** **(14 marks)**

Part A. A bullet of mass 20 g is fired horizontally at 400 m s^{-1} into a 2 kg wooden block hanging at rest, and becomes embedded in it.

1. Find the velocity of the block and bullet just after impact. (3)
2. Find the fraction of the initial kinetic energy lost in the impact, and state what be-

comes of this energy. (3)

Part B. A passenger of mass 70 kg is brought to rest from 20 m s^{-1} during a crash.

(c) Compute the average force on the passenger if the stopping time is 0.01 s (no airbag) and if it is 0.12 s (with airbag). (4)

(d) Using the impulse–momentum theorem, explain clearly why an airbag reduces injuries, and name one other safety device based on the same principle. (4)

EXERCISE 16 **Explosion, rocket and centre of mass** **(14 marks)**

Part A. A stationary shell of mass 12 kg explodes into two fragments of masses 5 kg and 7 kg. The 5 kg fragment moves off at 84 m s^{-1}.

1. Find the velocity of the 7 kg fragment. (3)

2. Find the total kinetic energy released by the explosion, and explain its origin. (3)

Part B. A rocket ejects exhaust gas at 2000 m s^{-1} relative to the rocket, at a rate of 150 kg s^{-1}.

(c) Calculate the thrust on the rocket. (2)

(d) Using momentum considerations, explain why the rocket accelerates even in vacuum, where there is nothing to “push against”. (3)

Part C. Three particles of masses 1 kg, 2 kg and 3 kg are placed at the points (0, 0), (4, 0) and (0, 3) metres respectively.

(e) Find the position of the centre of mass of the system. (2)

(f) Show that if the system is isolated, its centre of mass moves with constant velocity. (1)

11 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — Multiple choice: the nature of momentum**

1. Correct answer: (b) a vector $\vec{p} = m\vec{v}$, parallel to $\vec{v}$.
   Justification: Linear momentum is defined as the product of mass (a scalar) and velocity (a vector). Multiplying a vector by a positive scalar gives a vector with the same direction as the original vector, so $\vec{p}$ is parallel to $\vec{v}$. Options (a) and (d) describe scalars (kinetic energy is $K = \frac{1}{2}mv^2$), and (c) is incorrect because the momentum vector is parallel, not perpendicular, to the velocity.

2. Correct answer: (a) the 2 kg stone.
   Justification: Kinetic energy $K = \frac{1}{2}mv^2$ is the same for both stones. Let $m_1 = 2\,\text{kg}$, $m_2 = 1\,\text{kg}$. Then
   $$\frac{1}{2}m_1v_1^2 = \frac{1}{2}m_2v_2^2 \;\Rightarrow\; v_1 = v_2\sqrt{\frac{m_2}{m_1}} = \frac{v_2}{\sqrt{2}}.$$
   Momentum is $p = mv$, so
   $$p_1 = m_1v_1 = 2 \times \frac{v_2}{\sqrt{2}} = \sqrt{2}\,v_2, \qquad p_2 = m_2v_2 = v_2.$$
   Since $\sqrt{2} > 1$, the heavier stone has the greater momentum. (Equivalently, $p = \sqrt{2mK}$, so for equal K the larger mass has the larger momentum.)

SOLUTION **Exercise 2 — True or false**

1. True. If the net external force on a system is zero, the total momentum of the system is conserved (it remains constant in both magnitude and direction). This is the principle of conservation of linear momentum, which follows from $\vec{F}_{\text{ext}} = \frac{d\vec{P}}{dt} = \vec{0}$.

2. False. In a perfectly inelastic collision, the colliding bodies stick together and move with a common velocity after impact. Kinetic energy is not conserved; part of it is transformed into internal energy (heat, sound, permanent deformation). Only the total momentum is conserved.

3. True. Impulse $\vec{I} = \int_{t_1}^{t_2} \vec{F}\,dt$ is exactly the area under the force–time graph. For a constant force, $I = F\Delta t$ (area of a rectangle); for a varying force, the integral gives the area under the curve.

4. False. Before the explosion the body is stationary, so the total momentum is zero. By

conservation of momentum, the vector sum of the momenta of all the fragments after the explosion must also be zero. Therefore the fragments cannot all move in the same direction: their momenta must cancel, so for two fragments they move in opposite directions.

SOLUTION **Exercise 3 — Definitions**

1. Linear momentum of a particle: The linear momentum $\vec{p}$ of a particle of mass m moving with velocity $\vec{v}$ is defined by $\vec{p} = m\vec{v}$. It is a vector quantity, parallel to $\vec{v}$. SI unit: $\mathrm{kg\,m\,s^{-1}}$ (equivalently N s).

2. Impulse of a force: The impulse $\vec{I}$ of a force $\vec{F}$ acting over the time interval $[t_1, t_2]$ is defined by $\vec{I} = \int_{t_1}^{t_2} \vec{F}\,dt$. For a constant force, $\vec{I} = \vec{F}\,\Delta t$. By the impulse–momentum theorem, the impulse equals the change in momentum of the body: $\vec{I} = \Delta\vec{p}$. SI unit: N s (equivalently $\mathrm{kg\,m\,s^{-1}}$).

3. Elastic collision: A collision in which both the total momentum and the total kinetic energy of the system are conserved. The bodies do not stick together and no kinetic energy is converted into internal energy.

4. Centre of mass of a system of particles: The centre of mass is the point whose position vector is

$$\vec{R}_{\mathrm{cm}} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i},$$

where m_i and $\vec{r}_i$ are the masses and position vectors of the particles. The whole system moves as if its total mass were concentrated at this point and all the external forces acted there; in particular $\vec{P} = M\vec{v}_{\mathrm{cm}}$. SI unit of $\vec{R}_{\mathrm{cm}}$: metre (m).

SOLUTION **Exercise 4 — Direct calculations**

1. Momentum $p = mv = 8000 \times 15 = 120\,000\ \mathrm{kg\,m\,s^{-1}}$ (i.e. 1.2×10^5 N s). Direction: along the road, in the direction of motion.

2. Change in momentum of the ball (taking the initial direction of motion as positive):

$$\Delta p = m(v_f - v_i) = 0.058\,(0 - 30) = -1.74\ \mathrm{kg\,m\,s^{-1}}.$$

Magnitude of the impulse = 1.74 N s. Average force:

$$F_{\mathrm{avg}} = \frac{|\Delta p|}{\Delta t} = \frac{1.74}{0.05} = 34.8\ \mathrm{N},$$

directed opposite to the initial motion of the ball.

3. Impulse $I = F\,\Delta t = 40 \times 0.25 = 10$ N s, in the direction of the force.

SOLUTION **Exercise 5 — Kicking a football**

Take the initial direction of motion of the ball as positive.

1. Initial momentum: $p_i = 0.45 \times 18 = 8.1\,\text{kg m s}^{-1}$.
   Final momentum: $p_f = 0.45 \times (-24) = -10.8\,\text{kg m s}^{-1}$ (the ball moves back, hence the negative sign).
   Impulse: $I = \Delta p = p_f - p_i = -10.8 - 8.1 = -18.9$ N s.
   Magnitude = 18.9 N s; direction opposite to the initial motion of the ball.

2. Average force:
$$F_{\text{avg}} = \frac{I}{\Delta t} = \frac{-18.9}{0.08} = -236.25\ \text{N}.$$
   Magnitude ≈ 236 N; the negative sign shows that the force exerted by the foot on the ball acts opposite to the initial motion, i.e. in the direction of the final velocity.

SOLUTION **Exercise 6 — Man and canoe**

1. Take the direction of the man's jump as positive. Initially the man and the canoe are at rest, so the total momentum is 0. After the jump, conservation of momentum gives
$$m_{\text{man}}v_{\text{man}} + m_{\text{canoe}}v_{\text{canoe}} = 0 \;\Rightarrow\; 60 \times 3 + 90\,v_{\text{canoe}} = 0 \;\Rightarrow\; v_{\text{canoe}} = -\frac{180}{90} = -2\,\text{m s}^{-1}.$$
   The canoe recoils at $2\,\text{m s}^{-1}$ in the direction opposite to the man's jump.

2. Principle used: conservation of linear momentum. It applies because the man–canoe system is isolated in the horizontal direction: during the very short time of the jump, the external horizontal forces (water resistance) are negligible. The vertical forces (weights and buoyancy) balance each other and do not affect the horizontal momentum.

SOLUTION **Exercise 7 — Trolleys that stick together**

Take the direction of motion of the 2 kg trolley as positive.

1. Initial momentum:
$$p_i = (2 \times 5) + \big(3 \times (-1)\big) = 10 - 3 = 7\,\text{kg m s}^{-1}.$$
   After the collision the trolleys stick together: combined mass = 2 + 3 = 5 kg, common

velocity v. Conservation of momentum:

$$5v = 7 \;\Rightarrow\; v = 1.4\ \text{m s}^{-1},$$

in the positive direction (the direction initially taken by the 2 kg trolley).

2. Initial kinetic energy:

$$K_i = \tfrac{1}{2}(2)(5^2) + \tfrac{1}{2}(3)(1^2) = 25 + 1.5 = 26.5\ \text{J}.$$

Final kinetic energy:

$$K_f = \tfrac{1}{2}(5)(1.4^2) = 2.5 \times 1.96 = 4.9\ \text{J}.$$

Energy lost: $\Delta K = 26.5 - 4.9 = 21.6$ J.
Percentage lost:

$$\frac{21.6}{26.5} \times 100\% \approx 81.5\%.$$

This large loss is typical of a perfectly inelastic collision: the lost kinetic energy becomes heat, sound and deformation of the coupling.

SOLUTION **Exercise 8 — Force–time graph**

The graph is a triangle of base 0.4 s and height 500 N.

1. The impulse equals the area under the force–time graph:

$$I = \text{area of triangle} = \tfrac{1}{2} \times \text{base} \times \text{height} = \tfrac{1}{2} \times 0.4 \times 500 = 100\ \text{N s}.$$

2. By the impulse–momentum theorem, $I = \Delta p = m\,\Delta v$. For a 4 kg body initially at rest:

$$\Delta v = \frac{I}{m} = \frac{100}{4} = 25\ \text{m s}^{-1}.$$

The body reaches a velocity of $25\ \text{m s}^{-1}$ in the direction of the force.

SOLUTION **Exercise 9 — Collision at a crossroads**

Mark scheme and examiner expectations:

- (a) 2 marks: State the principle (conservation of linear momentum) and the condition (external forces negligible during the very short collision, so their impulse is approximately zero).
- (b) 3 marks: Correct components: east $p_x = 1500 \times 25 = 37\,500\ \text{kg m s}^{-1}$; north $p_y =$

$2000 \times 15 = 30\,000$ kg m s^{-1}.

- (c) 3 marks: Combined mass = 3500 kg; speed $v = \dfrac{\sqrt{37\,500^2 + 30\,000^2}}{3500} \approx 13.7$ m s^{-1}.
- (d) 2 marks: Angle $\theta = \arctan(30\,000/37\,500) \approx 38.7^\circ$ north of east.
- (e) 2 marks: KE lost $\approx 3.64 \times 10^5$ J (about 53% of the initial KE). Comment: perfectly inelastic collision (vehicles lock together), kinetic energy not conserved.

Detailed solution:

1. Principle: conservation of linear momentum, applied component by component. It is valid because during the very short collision time the external forces (friction, air resistance) are negligible compared with the large internal forces between the vehicles, so the impulse of the external forces is approximately zero.

2. Momentum components before the collision:
East (x-direction): $p_x = m_{\text{car}} v_{\text{car}} = 1500 \times 25 = 37\,500$ kg m s^{-1}.
North (y-direction): $p_y = m_{\text{truck}} v_{\text{truck}} = 2000 \times 15 = 30\,000$ kg m s^{-1}.

3. Speed after the collision: combined mass $M = 1500 + 2000 = 3500$ kg. Momentum is conserved in each direction:

$$Mv_x = p_x \Rightarrow v_x = \frac{37\,500}{3500} = 10.71 \text{ m s}^{-1}, \qquad Mv_y = p_y \Rightarrow v_y = \frac{30\,000}{3500} = 8.57 \text{ m s}^{-1}.$$

Hence

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{10.71^2 + 8.57^2} = \sqrt{188.3} \approx 13.7 \text{ m s}^{-1}.$$

4. Direction:

$$\theta = \arctan\left(\frac{v_y}{v_x}\right) = \arctan\left(\frac{8.57}{10.71}\right) = \arctan(0.8) \approx 38.7^\circ \quad \text{north of east.}$$

5. Kinetic energy lost:

$$K_i = \frac{1}{2}(1500)(25^2) + \frac{1}{2}(2000)(15^2) = 468\,750 + 225\,000 = 693\,750 \text{ J},$$

$$K_f = \frac{1}{2}(3500)(13.7^2) \approx 328\,500 \text{ J},$$

$$\Delta K = K_i - K_f \approx 693\,750 - 328\,500 \approx 3.65 \times 10^5 \text{ J} \quad (\text{about } 53\% \text{ of } K_i).$$

Comment: the collision is perfectly inelastic (the vehicles lock together), so kinetic energy is not conserved; the lost energy is transformed into heat, sound and the work of deforming the vehicles.

SOLUTION **Exercise 10 — Ballistic pendulum and airbag safety**

Mark scheme and examiner expectations:

- Part A (a) 3 marks: Conservation of momentum: $0.02 \times 400 = 2.02\,v \Rightarrow v \approx 3.96\ \text{m s}^{-1}$.
- Part A (b) 3 marks: $K_i = 1600$ J, $K_f \approx 15.8$ J; fraction lost ≈ 0.99. Energy becomes heat, sound and deformation.
- Part B (c) 4 marks: Without airbag: $F = \dfrac{70 \times 20}{0.01} = 140\,000$ N. With airbag: $F = \dfrac{1400}{0.12} \approx 11\,700$ N.
- Part B (d) 4 marks: Impulse–momentum theorem: Δp fixed; increasing Δt reduces the average force. Other device: seat belt, crumple zone or crash helmet.

Detailed solution:

Part A.

1. Let $m_b = 0.020$ kg, $u_b = 400\ \text{m s}^{-1}$, $M = 2.00$ kg (block initially at rest). After the bullet embeds itself, the combined mass is $M + m_b = 2.02$ kg moving with velocity v. Conservation of momentum:

$$m_b u_b = (M + m_b)v \;\Rightarrow\; v = \frac{0.02 \times 400}{2.02} = \frac{8}{2.02} \approx 3.96\ \text{m s}^{-1}.$$

2. Initial kinetic energy:

$$K_i = \frac{1}{2} m_b u_b^2 = \frac{1}{2} \times 0.02 \times 400^2 = 1600\ \text{J}.$$

Final kinetic energy:

$$K_f = \frac{1}{2}(2.02)(3.96^2) \approx 15.8\ \text{J}.$$

Fraction lost:

$$\frac{K_i - K_f}{K_i} = \frac{1600 - 15.8}{1600} \approx 0.990 \quad \text{(i.e. about 99\%)}.$$

The lost kinetic energy is converted into heat, sound and the work done in deforming the block and the bullet (internal energy).

Part B.

(c) Passenger mass $m = 70$ kg, initial speed $u = 20\ \text{m s}^{-1}$, final speed 0. The magnitude of the change of momentum is

$$|\Delta p| = m\,|\Delta v| = 70 \times 20 = 1400\ \text{kg m s}^{-1}.$$

Without airbag ($\Delta t = 0.01$ s):

$$F_{\text{avg}} = \frac{|\Delta p|}{\Delta t} = \frac{1400}{0.01} = 140\,000\ \text{N}.$$

With airbag ($\Delta t = 0.12$ s):

$$F_{\text{avg}} = \frac{1400}{0.12} \approx 11\,700\ \text{N}.$$

(d) The impulse–momentum theorem states $F_{\text{avg}}\,\Delta t = \Delta p$. For a given crash, the change of momentum Δp of the passenger is fixed by the initial speed. By increasing the stopping time Δt (from 0.01 s to 0.12 s, a factor of 12), the airbag divides the average force on the passenger by 12. This much smaller force reduces the pressure on the body and therefore the risk of serious injury. Another safety device based on the same principle is the crumple zone of a car (also seat belts and crash helmets), which lengthens the time over which the momentum of the occupants changes.

SOLUTION **Exercise 11 — Explosion, rocket and centre of mass**

Mark scheme and examiner expectations:

- Part A (a) 3 marks: Conservation of momentum: $0 = 5 \times 84 + 7v \Rightarrow v = -60\ \text{m s}^{-1}$.
- Part A (b) 3 marks: KE released = $\frac{1}{2}(5)(84^2) + \frac{1}{2}(7)(60^2) = 30\,240$ J. Origin: chemical potential energy of the explosive.
- Part B (c) 2 marks: Thrust = $v_{\text{rel}}\dfrac{dm}{dt} = 2000 \times 150 = 300\,000$ N.
- Part B (d) 3 marks: The rocket pushes the exhaust backward; by Newton's third law the exhaust pushes the rocket forward. No external medium is needed; momentum of the rocket + exhaust system is conserved.
- Part C (e) 2 marks: $x_{\text{cm}} = \dfrac{8}{6} \approx 1.33$ m; $y_{\text{cm}} = \dfrac{9}{6} = 1.5$ m.
- Part C (f) 1 mark: $\vec{F}_{\text{ext}} = \vec{0} \Rightarrow \dfrac{d\vec{P}}{dt} = \vec{0} \Rightarrow \vec{P}$ = constant; since $\vec{P} = M\vec{v}_{\text{cm}}$, $\vec{v}_{\text{cm}}$ is constant.

Detailed solution:

Part A.

1. The shell is initially at rest, so the total momentum before the explosion is 0. Take the direction of the 5 kg fragment as positive and let v be the velocity of the 7 kg fragment. Conservation of momentum:

$$5 \times 84 + 7v = 0 \;\Rightarrow\; v = -\frac{420}{7} = -60\ \text{m s}^{-1}.$$

   The 7 kg fragment moves at $60\ \text{m s}^{-1}$ in the direction opposite to the 5 kg fragment.

2. Kinetic energy after the explosion:

$$K = \frac{1}{2}(5)(84^2) + \frac{1}{2}(7)(60^2) = \frac{1}{2}(35\,280 + 25\,200) = 30\,240\ \text{J}.$$

   Since the initial kinetic energy was zero, the energy released by the explosion is 30 240 J. It comes from the chemical potential energy stored in the explosive, converted into kinetic energy of the fragments.

Part B.

(c) The thrust is the rate of change of momentum of the exhaust gas:

$$F = v_{\text{rel}} \times \frac{dm}{dt} = 2000 \times 150 = 300\,000\ \text{N} = 3.0 \times 10^5\ \text{N}.$$

(d) The rocket accelerates by ejecting mass (exhaust gas) backwards at high speed relative to itself. The rocket and the exhaust gas form an isolated system, so their total momentum is conserved: the forward momentum gained by the rocket is equal in magnitude to the backward momentum carried away by the exhaust. Equivalently, the rocket exerts a backward force on the gas, and by Newton's third law the gas exerts a forward reaction force (the thrust) on the rocket. No external medium (such as air) is required, which is why rockets work in the vacuum of space.

Part C.

(e) Total mass $M = 1 + 2 + 3 = 6$ kg. The coordinates of the centre of mass are

$$x_{\text{cm}} = \frac{1(0) + 2(4) + 3(0)}{6} = \frac{8}{6} = \frac{4}{3} \approx 1.33\ \text{m},$$

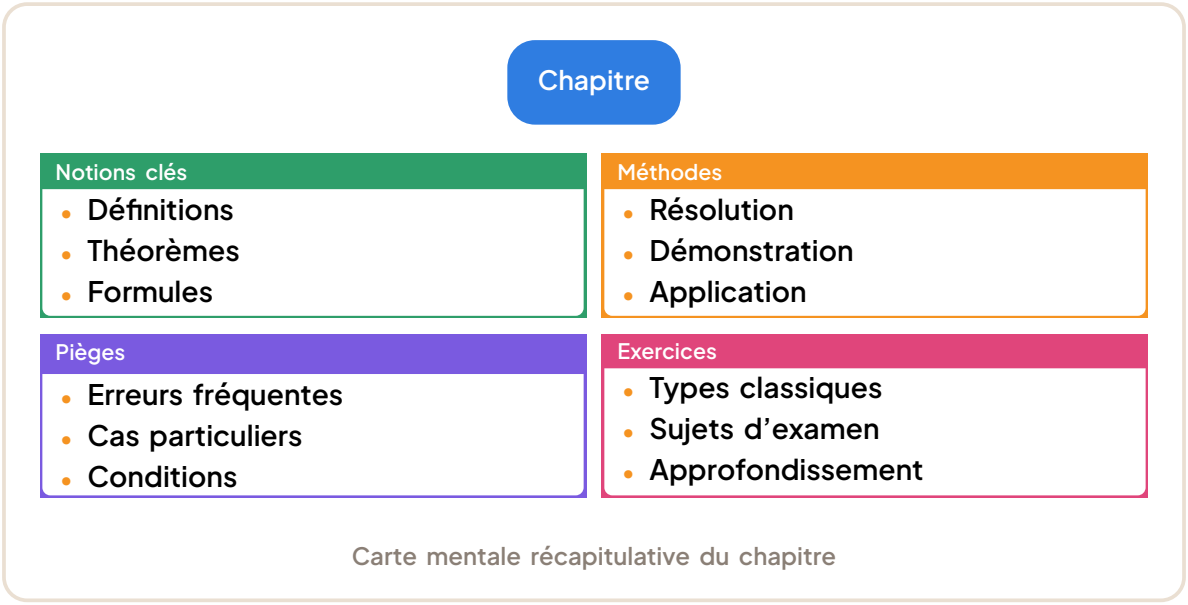
$$y_{\text{cm}} = \frac{1(0) + 2(0) + 3(3)}{6} = \frac{9}{6} = 1.5\ \text{m}.$$

The centre of mass is at the point $\left(\frac{4}{3},\ 1.5\right)$ m.

(f) For an isolated system the net external force is zero: $\vec{F}_{\text{ext}} = \vec{0}$. Newton's second law for a system of particles gives $\vec{F}_{\text{ext}} = \frac{d\vec{P}}{dt}$, where $\vec{P}$ is the total momentum. Hence $\frac{d\vec{P}}{dt} = \vec{0}$, so $\vec{P}$ is constant. But $\vec{P} = M\vec{v}_{\text{cm}}$ with constant total mass M, therefore $\vec{v}_{\text{cm}}$ is constant: the centre of mass moves with constant velocity (or remains at rest if it was initially at rest).

12 SUMMARY SHEET

KEY POINTS **Bilan du chapitre**



Carte mentale récapitulative du chapitre

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